PromptHSI-CD: A Land-Cover Category Prompt-Guided Method for Hyperspectral Change Detection

Weipeng Jing, *Member, IEEE*, Linkang Geng, Ye Yuan, Guangliang Cheng, Fangyu Sun, and Chao Li

***Abstract*—Hyperspectral images, with their richer spectral information than multispectral images, are increasingly used for identifying land-cover changes. Deep-learning-based hyperspectral image change detection (HSI-CD) has gained significant attention, and many methods use carefully designed feature extractors for bi-temporal HSI. Most still pursue a pixel-to-binary change map inference pathway that demands abundant labels. Given the high cost of annotation and the scarcity of labeled samples, models are sensitive to training data size and vulnerable to distribution shift, which limits performance in few-shot scenarios. Approaches that claim few-shot capability often require pretraining on very-high-resolution source-domain data, which does not meaningfully reduce data acquisition and annotation costs. To address these challenges, we propose a few-shot multimodal HSI-CD method with predefined land-cover category prompts, namely PromptHSI-CD. The fixed prompts introduce stable category-level semantic priors and transform the inference pathway from direct pixel-to-binary mapping to a land-cover feature-mediated mapping that produces the binary change map, which mitigates the impact of limited pixel samples and distribution shift under few-shot scenarios. We further design a semantic-guided feature fusion (SGFF) module to align global text semantics with bi-temporal visual features, and a multiperspective change-aware feature enhancement (CAFE) module with multimodal-guided feature attention to improve sensitivity to changed regions. Extensive experiments on five widely used datasets show that the proposed method outperforms state-of-the-art (SOTA) baselines. In particular, under severe class imbalance in the River dataset and with only 0.1% labeled samples, PromptHSI-CD achieves an $F1$-score of 0.6883 and a Kappa coefficient of 0.6590, surpassing the SOTA by 5.48% and 4.9%, respectively. The code is available at https://github.com/ylfbx329/PromptHSI-CD**

***Index Terms*—Change detection, change-aware enhancement, hyperspectral image, multimodal network.**

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I. Introduction

REMOTE sensing change detection (RSCD) aims to identify changes in the same geographical region across different time points [1]. It plays a crucial role in remote sensing applications, such as urban development assessment [2], [3], forest monitoring [4], [5], and natural disaster detection [6], [7]. Compared with common RGB and multispectral images that contain only 3–12 bands, hyperspectral images comprise 150–250 bands and provide much richer spectral information, enabling more accurate discrimination of land-cover categories [8], [9], [10]. Therefore, hyperspectral image change detection (HSI-CD) can achieve more refined identification of land-cover changes, offering particular advantages in areas such as farmland and water bodies that exhibit distinctive spectral response characteristics [11].

Traditional HSI-CD methods obtain differential features by performing algebraic operations (e.g., subtraction) on bi-temporal images and then distinguish change regions using adaptive thresholding [12] or clustering methods [13]. With the development of remote sensing technology and the rise of deep learning, applying deep learning approaches to HSI-CD has become a research hotspot [14], [15], [16]. In the early stages, Wang et al. [17] proposed GETNET, a general end-to-end 2-D CNN framework that leveraged spectral unmixing to extract subpixel endmember abundance information. It constructed a mixed-affinity matrix from hyperspectral images and their corresponding abundance maps, which was then fed into a CNN to classify whether a pixel has changed. Luo et al. [18] designed a simple U-Net-like [19] multiscale encoder–decoder baseline and integrated attention modules to optimize differential feature extraction. However, traditional algebraic approaches are highly sensitive to hyperparameters such as thresholds and cluster size, and their settings often rely on manual adjustment, which hampers generalization on large-scale data. In addition, the manually constructed mixed-affinity matrix may limit the model's ability to represent more complex spectral variations, and the relatively simple CNN architecture is often insufficient to capture complex change patterns in bi-temporal images.

In recent studies, a number of advanced end-to-end models have been proposed and applied to HSI-CD tasks. Zhang et al. [20] addressed affine misregistration in bi-temporal hyperspectral images by exploiting SAM-generated object-level masks for structurally consistent registration and integrating region-

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based label propagation with a spatial–spectral–temporal modeling network. Qu et al. [21] proposed MambaFedCD, a spatial–spectral–temporal collaborative Mamba-based active federated framework for HSI-CD, which enabled change detection across multiple clients without sharing raw data while reducing the manual labeling burden. In a related direction, Cao et al. [22] developed RMNet for remote sensing image change captioning, which combined the CNN and VMamba in a dual-stream feature extraction framework and introduced a spatial- and channel-wise difference recalibration mechanism to emphasize real changes and suppress pseudo-changes, thereby improving the quality of change-aware semantic descriptions. Teymoor Seydi et al. [23] compared the performance of five different Kolmogorov–Arnold networks (KANs) on HSI-CD tasks. The KAN is based on the Kolmogorov–Arnold representation theorem [24], which states that any continuous multivariate function can be represented as a composition of continuous univariate functions. Compared with a multilayer perceptron (MLP), KANs reduce the number of parameters and improve computational efficiency. However, due to the high cost of acquiring hyperspectral image data, it is difficult to obtain large-scale labeled datasets, which fundamentally conflicts with the requirement of deep learning models for large-scale training data, thereby limiting models' ability to generalize in image feature learning [25].

As the number of training samples decreases, these models fail to learn a reliable inference paradigm from image features to binary change maps under few-shot conditions, leading to a rapid deterioration in performance. In contrast, Hou et al. [26] proposed a cross-scene few-shot learning method based on source–target-domain adaptation. By integrating a contrastive learning strategy, they pretrain the model using very-high-resolution images as source-domain data and subsequently fine-tune it on a small set of target-domain hyperspectral samples, while also designing a cross-domain contrastive alignment module for feature alignment, thereby achieving few-shot learning. Although this approach alleviates the dependence on labeled hyperspectral samples to some extent, it requires an additional very-high-resolution image as source-domain data, which increases the difficulty of data acquisition and does not substantially reduce the overall cost of data collection and annotation.

With the rise of multimodal learning, exploiting multimodal data to address change detection has become a promising direction [27]. Numerous studies report the superior performance and generalization of multimodal models in specific scenarios [28]. One stream of research focuses on foundational multimodal modeling, introducing general-purpose models such as CLIP [29], Florence [30], and Flamingo [31]. By jointly learning visual and textual representations during pretraining, these models acquire richer and more universal semantics. Visual features align with natural language supervision, open vocabulary knowledge is distilled from web-scale corpora, and cross-domain transferability improves. As a result, they achieve few-shot and even zero-shot performance on downstream tasks. Other work explores task-specific solution spaces using multimodal priors. For RGB remote sensing image change detection, Dong et al. [32] proposed ChangeCLIP, which adopted a two-stage classification-then-detection paradigm. It first employed CLIP to classify the RGB images and then fed both the classification results and the original RGB images into ChangeCLIP for feature extraction and subsequent change detection. Overall, the evidence shows that pretrained language models encode valuable implicit knowledge that benefits remote sensing tasks. Given that annotating hyperspectral data is substantially more costly than labeling conventional multispectral or RGB images, the text modality can serve as an effective semantic before assist visual representation learning in few-shot scenarios.

In the HSI-CD task, hyperspectral images typically contain 150–250 spectral bands, many of which are redundant, making it difficult to directly learn discriminative representations for binary change detection. In contrast, textual data describing land-cover categories naturally carry semantic priors that are beneficial for distinguishing meaningful changes. Therefore, we introduce predefined land-cover category information into the HSI-CD framework to provide semantic guidance. As illustrated in Fig. 1, this design transforms the inference pathway from the traditional pixel-to-binary change map paradigm into a pixel-to-land-cover category probability distribution-to-binary change map paradigm, thereby aligning the detection process with the intrinsic nature of land-cover transitions under few-shot scenarios. The main contributions of this article are summarized as follows.

1) We propose PromptHSI-CD, a multimodal HSI-CD method that incorporates predefined land-cover prompts and constructs a semantically guided inference pathway from pixels to the change map through land-cover category probability distributions. By introducing transferable land-cover semantics, the method reduces reliance on low-level spectral change patterns that are difficult to learn robustly from limited labeled samples, thereby achieving superior performance under limited training samples.
2) We develop a semantic-guided feature fusion (SGFF) module that integrates textual land-cover embeddings with bi-temporal hyperspectral features to build a more stable semantic intermediate representation. By injecting category-level semantic information into visual features, SGFF reduces the model's dependence on spectral change patterns learned from limited samples, patterns that are often incomplete and insufficiently representative of the overall change patterns in the data under limited training samples, thereby reducing overfitting and improving generalizability.
3) We design a change-aware feature enhancement (CAFE) module that enhances sensitivity to semantic and structural differences between temporal features. Through multiperspective enhancement and multimodal-guided feature attention, CAFE captures complementary change features from spatial, frequency, semantic, and contextual perspectives, thereby enabling more robust and generalizable change feature extraction under limited training samples.

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4) We conduct comprehensive experiments on five benchmark datasets. Both quantitative and qualitative results demonstrate that PromptHSI-CD achieves superior performance compared with state-of-the-art (SOTA) methods, particularly under few-shot scenarios.

The remainder of this article is organized as follows. Section II reviews the related work. Section III presents the overall architecture of PromptHSI-CD and details its key modules. Section IV describes the experimental setup and discusses the results. Section V discusses the limitations of the proposed method and future research directions. Finally, Section VI concludes this article.

II. Related Work

A. Deep Learning for HSI-CD

Recent deep learning methods for HSI-CD have mainly focused on improving spectral–spatial feature representations and enhancing the modeling of bi-temporal changes. Wang et al. [33] proposed SFFGL, which used a global spatial-wise FCN with a spatial attention mechanism to achieve patch-free inference and stronger global context exploitation. Jiang et al. [34] proposed ACFHAN, which combined adaptive spatial–spectral hybrid self-attention with center-focused attention to enhance the area most relevant to the center pixel. Another important direction is to strengthen differential feature extraction and bi-temporal interaction. Wu et al. [35] focused on differential features, leveraging attention mechanisms and the principle of maximum differential entropy to capture change information, along with gated logic for feature filtering. Qin et al. [36] proposed a network structure that aligns the means and variances of bi-temporal images, alleviating the inconsistency of spatial feature distributions.

Besides fully supervised learning, reducing the dependence on dense annotations has become another major theme in HSI-CD. Zhou et al. [37] designed manually crafted masks for single-temporal images and developed an algorithm to generate corresponding masks for the second temporal phase, enabling self-supervised learning. Jian et al. [38] proposed a hypergraph-based self-supervised learning method for joint spatial–temporal–spectral feature representations. Although these methods improve label efficiency and representation quality, they still remain within the image modality and mainly focus on visual feature learning itself, without explicitly introducing external semantic priors to guide the HSI-CD process.

However, most existing methods still follow a pixel-to-binary mapping paradigm that either depends on sufficient supervision or resorts to self-supervised learning to capture complex spectral change patterns. In contrast, we propose PromptHSI-CD, a multimodal HSI-CD framework that provides stable semantic guidance through predefined land-cover prompts independent of the input image, explicitly aligns global textual semantics with bi-temporal visual features through the SGFF module, and strengthens change-relevant representations from multiple perspectives through the CAFE module. As a result, this end-to-end multimodal design maintains strong and consistent performance in few-shot scenarios across datasets of varying complexity.

B. Multimodal Methods in RSCD

With the advancement of artificial intelligence, the integration between computer vision (CV) and natural language processing (NLP) has accelerated the development of multimodal learning. By jointly modeling image and text modalities, multimodal models are capable of learning more general and semantically enriched representations, improving performance across a wide range of downstream tasks. Notable examples include CLIP [29], which is trained on 400 million image–text pairs via contrastive learning to construct a unified vision–language embedding space, enabling cross-modal generalization in tasks such as zero-shot classification and image–text retrieval. BLIP-2 [39] further improves visual–language alignment and supports more complex tasks like visual question answering.

Recently, such models have begun to be applied to remote sensing. For instance, ChangeCLIP [32] adopts a two-stage classification-then-detection strategy, where a pretrained CLIP model is first used to classify RGB remote sensing images and the resulting predictions are incorporated into the subsequent change detection stage. However, ChangeCLIP is not well suited for HSI-CD for two main reasons. First, CLIP is pretrained exclusively on RGB images and therefore cannot directly classify hyperspectral data. Compared with RGB images, hyperspectral data typically have lower spatial resolution and public HSI datasets rarely provide textual descriptions of land-cover categories; as a result, forcing CLIP to operate on the RGB channels of hyperspectral images yields low-confidence predictions. Second, the two-stage design of ChangeCLIP introduces an inherent error propagation problem, whereby inaccuracies in the classification stage inevitably degrade the downstream detection performance. These limitations highlight the need for an end-to-end multimodal method specifically designed for HSI-CD. M^3amba [40], which targets hyperspectral multimodal land-cover classification, uses a pretrained CLIP visual encoder with modality-specific adapters to extract modality-aware semantic features from HSI together with SAR or LiDAR. These CLIP-driven adapters alleviate modality mismatch and make the original CLIP encoder modality-aware. The extracted features are then fused using a linear-complexity Mamba network equipped with a cross-attention module to enforce cross-modal consistency and exploit complementary information.

However, multimodal models for HSI-CD remain nascent and face two key challenges. First, most existing architectures are designed for natural images and transfer poorly to hyperspectral images, which exhibit high spectral dimensionality and bi-temporal variability. Second, current approaches often use text as a coarse global hint, lacking detailed cross-modal fusion and explicit change-aware modeling. To address these issues, we develop an end-to-end multimodal HSI-CD method that conditions hyperspectral spectral–spatial features on category-level land-cover prompts and performs fine-grained cross-modal fusion with explicit change-aware reasoning. Fixed prompts are embedded by a pretrained text encoder to provide stable semantic anchors; these anchors guide the refinement of multiscale hyperspectral

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TABLE I
56 LAND-COVER CATEGORY PROMPTS DEFINED IN [32] USED AS TEXT INPUTS TO PROMPTHSI-CD

Group	Categories
Nature	Beach, Forest, Lake, Meadow, Mountain, Sea, Wetland, Cotton Field, Farmland, Prairie, Desert, River, Tree, Shrubbery, Chaparral, Fertile Land, Snow Land, Pond, Island
Transport	Airport, Bridge, Freeway, Harbor, Railway, Interchange, Intersection, Road, Highway
Sports	Basketball Court, Ground Track Field, Stadium, Tennis Court, Golf Course
Residential	Dense Residential, Single-Family Residential, Building, Church, Cabin
Industrial	Commercial Area, Industrial Area, Oil Tank, Storage Tanks, Container, Mine
Other	Terrace, Campus, Park, Parking Lot, Square, Solar Panel, Cars, Ship, Airplane, Runway, Impermeable Surface

representations and the bi-temporal differencing process. This design adapts natural image priors to hyperspectral data and replaces coarse global text cues with targeted, semantic-aware change localization.

III. METHODOLOGY

This section introduces the overall architecture of the proposed land-cover category prompt-guided HSI-CD method, PromptHSI-CD, and details its key modules, including multimodal feature extraction, SGFF, CAFE, and differential feature decoder.

A. Overall Architecture

PromptHSI-CD introduces predefined land-cover category prompts to guide change detection and therefore requires effective matching between hyperspectral images and prompts. However, the direct CLIP-based matching strategy adopted in previous work [32] is unsuitable for HSI, and its multistage design may further lead to error accumulation when the visual–language prior is not well matched to hyperspectral data. To address these issues, we modify the patch embedding layer to support hyperspectral inputs and design an adaptive prompt selection and fusion mechanism, thereby integrating the preclassification stage into the model itself and enabling end-to-end inference. In this way, prompt semantics can be directly aligned with HSI visual features within a unified framework, avoiding the adverse effects of inaccurate one-shot classification on subsequent change detection.

Moreover, unlike single-modal change detection approaches that rely solely on bi-temporal image inputs, PromptHSI-CD additionally incorporates a text prompt comprising 56 predefined land-cover categories (see Table I). By introducing transferable land-cover semantics, the proposed method reduces reliance on low-level spectral change patterns that are difficult to learn robustly from limited labeled samples, thereby improving representation learning under few-shot supervision. Through this end-to-end multimodal design, PromptHSI-CD achieves more robust and generalizable change representation learning, which is particularly beneficial in few-shot scenarios across datasets with varying complexity.

As shown in Fig. 2, the overall architecture comprises four components: a multimodal encoder, SGFF, CAFE, and a differential feature decoder. The main implementation steps are summarized in Algorithm 1.

1) *Multimodal Encoder:* Two weight-sharing visual encoders extract multiscale features from the bi-temporal images. A text encoder converts predefined land-cover category prompts into embeddings that provide auxiliary semantic guidance for model inference.
2) *SGFF:* This module selects and enhances class-specific text embeddings and aligns and fuses them with the visual features. The resulting semantic intermediate representation, enriched with category-level semantic information, reduces reliance on the incomplete or unstable spectral change patterns often encountered in few-shot scenarios, thereby mitigating overfitting and improving representation robustness.
3) *CAFE:* This module enhances bi-temporal differences through multiview differencing on multiscale visual features and category-aware reweighting with multimodal features. This design enables more reliable extraction of discriminative change patterns from limited training samples, thereby reducing sensitivity to unstable variations and improving robustness and generalization in few-shot scenarios.
4) *Decoder:* The decoder refines the aggregated difference features and outputs change maps consistent with the requirement of the HSI-CD task.

B. Multimodal Feature Extraction

To incorporate land-cover semantics into the HSI-CD task, we construct a set of land-cover category prompts to guide the model in identifying land-cover changes. Specifically, we design two types of prompt templates: foreground class prompts in the form of "Remote sensing image patch for change detection: Class Name" and a background class prompt in the form of "Remote sensing image patch for change detection: background" [32] compiled a list of 56 land-cover categories commonly observed in remote sensing images, as shown in Table I, and we adopt these as our class names. Using a foreground prompt template filled with each of the 56 categories and adding one background prompt, we construct 57 category-specific prompt sentences, which are used as inputs to the text encoder.

Following previous CLIP-based studies in the hyperspectral field [41], [42], [43], we use a frozen CLIP text encoder to extract semantic representations from the prompts to preserve its strong textual semantic extraction capability acquired during pretraining. In few-shot scenarios, freezing the text encoder also helps stabilize training and maintain consistent semantic grounding.

For bi-temporal hyperspectral images, we adopt a standard dual-stream architecture [34], [44], [45] using two weight-sharing visual encoders to extract features. Specifically, we

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Algorithm 1 PromptHSI-CD

Input: Bi-temporal images $\mathbf{I}_{T1}, \mathbf{I}_{T2} \in \mathbb{R}^{H\times W\times C}$; prompt word set $T_w = \{w_i\}_{i=1}^{M}$ ($M = 56$, see Table I); Top-k selection size k.
Output: Binary change map $Y \in \{0,1\}^{H\times W}$
// Step 0. Data Preparation
$T_s \leftarrow$ {remote sensing image patch for change detection: w_i. | $w_i \in T_w$}
Append "remote sensing image patch for change detection: background." to T_s
Divide $\mathbf{I}_{T1}$ and $\mathbf{I}_{T2}$ into patches
for each temporal image $\mathbf{I}$ **do**
  // Step 1. Multimodal feature extraction
  $\tilde{\mathbf{X}}_{text} \leftarrow$ text_embedding(tokenize(T_s))
  $\{\mathbf{I}, \tilde{\mathbf{X}}^i_{vis}, \mathbf{X}_{cls}\} \leftarrow$ vision_embedding($\mathbf{I}$), $i \in \{1,2,3,4\}$ ▷ multiscale visual features
  $\mathbf{X}_{vis} = [\mathbf{X}^i_{vis}, i \in \{1,2,3,4,5\}] \leftarrow$ FPN($\mathbf{I}, \tilde{\mathbf{X}}_{vis}$) ▷ cf. (1)
  // Step 2. Semantic-guided feature fusion
  $\mathbf{S}_{dot} \leftarrow$ Similarity($\mathbf{X}_{cls}, \tilde{\mathbf{X}}_{text}$) ▷ cf. (2)
  $\mathbf{X}_{text}, \mathbf{S}_{dot} \leftarrow$ Top-k($\tilde{\mathbf{X}}_{text}; \mathbf{S}_{dot}, k$) ▷ select text prompts by dot-product similarity $\mathbf{S}_{dot}$
  $\mathbf{S}_{KL} \leftarrow D_{KL}(\mathbf{X}_{cls} \,\|\, \mathbf{X}_{text})$ ▷ cf. (3)
  $\mathbf{X}_{cross} \leftarrow$ CrossAttn($\mathbf{X}^5_{vis}, \mathbf{X}_{text}$) ▷ cf. (4)
  $\mathbf{X}_{vt} \leftarrow$ Fuse($\mathbf{S}_{dot}, \mathbf{S}_{KL}, \mathbf{X}_{cross}, \mathbf{X}_{text}$) ▷ cf. (5)
  $\mathbf{X}_{score_map} \leftarrow \mathbf{X}_{vt} \times \mathbf{X}^5_{vis}$
  $\mathbf{X}^5_{vis} \leftarrow$ Concat($\mathbf{X}^5_{vis}, \mathbf{X}_{score_map}$) ▷ cf. (5)
end for
// Step 3. Change-aware feature enhancement
for each scale $i \in \{1,2,3,4,5\}$ **do**
  $\Delta\mathbf{F}_{sub} \leftarrow \text{Diff}_{spatial}(\mathbf{X}^{T1}_{vis}, \mathbf{X}^{T2}_{vis})$ ▷ cf. (7)
  $\Delta\mathbf{F}_{freq} \leftarrow \text{Diff}_{freq}(\mathbf{X}^{T1}_{vis}, \mathbf{X}^{T2}_{vis})$ ▷ cf. (8, 9, 10, 11, 12, 13)
  $\Delta\mathbf{F}_{fpn} \leftarrow$ FPN(Concat($(\mathbf{X}^i_{vis})^{T1}, (\mathbf{X}^i_{vis})^{T2}$))
  $\Delta\mathbf{F}_{cos} \leftarrow \text{Diff}_{cos}(\mathbf{X}^{T1}_{vis}, \mathbf{X}^{T2}_{vis}) \odot \Delta\mathbf{F}_{fpn}$ ▷ cf. (15, 16)
  $\Delta\mathbf{F}_{all} \leftarrow$ Concat($\Delta\mathbf{F}_{sub}, \Delta\mathbf{F}_{freq}, \Delta\mathbf{F}_{fpn}, \Delta\mathbf{F}_{cos}$)
  $\Delta\mathbf{F}^i_{change} \leftarrow$ ChangeAttn($\Delta\mathbf{F}_{all}, \mathbf{X}^{T1}_{vt}, \mathbf{X}^{T2}_{vt}$) ▷ cf. (17, 18)
end for
// Step 4. Differential feature decoding
$\Delta\mathbf{F} \leftarrow$ Concat($\Delta\mathbf{F}^i_{change}$), $i \in \{1,2,3,4,5\}$ ▷ Concatenate multiscale change features
$Y \leftarrow$ Decoder($\Delta\mathbf{F}$) ▷ stacked convs / activations

use the CLIP visual encoder and extract multiscale features from transformer layers at indices $[3,5,7,11]$, where the indices start from 0 and the total number of layers is 12. To better highlight the effectiveness of the proposed SGFF and CAFE modules, and to alleviate the risk of overfitting under the extremely limited supervision setting, we keep the CLIP vision encoder frozen. Only the patch embedding layer is modified to accommodate the hyperspectral input and is trained accordingly.

After encoding, each temporal input produces the original image, four multiscale visual features, and a class token, that is, $\{\mathbf{I}, \tilde{\mathbf{X}}^i_{vis}, \mathbf{X}_{cls} \mid i \in \{1,2,3,4\}\}$. Lower layers retain more spatial details, whereas higher layers capture more abstract semantics. To integrate multiscale information, we adopt a top-down fusion strategy based on feature pyramid networks (FPNs) [46] to fuse the visual features $\tilde{\mathbf{X}}'_{vis} = \{\mathbf{I}, \tilde{\mathbf{X}}^i_{vis} \mid i \in \{1,2,3,4\}\}$. As the encoder operates in a bottom-up manner, higher layers capture more abstract and semantically meaningful representations. By progressively upsampling and fusing features from top to bottom, we enhance the representation capability of lower layers across scales. The fusion is performed as follows:

$$\mathbf{X}^i_{vis} = \text{Conv}_{3\times3}\left(\text{Conv}_{1\times1}\left(\tilde{\mathbf{X}}'^i_{vis}\right) + \text{Upsample}\left(\text{Conv}_{1\times1}\left(\tilde{\mathbf{X}}'^{(i+1)}_{vis}\right)\right)\right) \quad i \in \{1,2,3,4\}. \tag{1}$$

As the fusion process is identical for both time steps, we use $\mathbf{X}^i_{vis}$ to represent features at scale i. In particular, we set $\mathbf{X}^5_{vis} = \tilde{\mathbf{X}}^4_{vis}$. $\text{Conv}_{1\times1}$ and $\text{Conv}_{3\times3}$ are convolution layers, and Upsample($\cdot$) denotes bilinear interpolation. The result is two sets of multiscale features with the same resolution and richer information compared with the original visual features.

Through multimodal feature extraction, we obtain multiscale visual features and a semantic representation of text prompts covering 56 land-cover categories plus background. After fusion with FPN, the visual features at different scales contain rich texture and semantic information, while the text prompt features capture general semantics of land-cover categories. We then apply class-specific selection and enhancement, followed by full fusion with the visual features.

C. Semantic-Guided Feature Fusion

After processing with the text and visual encoders, we obtain the text embedding $\tilde{\mathbf{X}}_{text} \in \mathbb{R}^{L\times d}$ and the multiscale visual embeddings $\{(\mathbf{X}^i_{vis})^t \in \mathbb{R}^{C\times H\times W}, \mathbf{X}_{cls} \mid i \in \{1,2,3,4,5\}, t \in \{T1,T2\}\}$, where $L = 57$ denotes the number of text prompts and d denotes the text feature dimension, which is set equal to the number of visual feature channels C to avoid unnecessary dimensional transformations. To address the challenge of learning reliable change representations from limited labeled samples, we design the SGFF module to construct a more stable semantic intermediate representation by selectively introducing land-cover semantics into the visual features. This semantic guidance compensates for the insufficiency of few-shot supervision, enabling the model to rely less on unstable low-level spectral change patterns and to achieve better generalization under distribution shift.

Specifically, we select $\mathbf{X}_{cls}$, which represents the most abstract semantic level of the visual embeddings, and compute scaled dot-product similarity with the text embedding $\tilde{\mathbf{X}}_{text}$. This enables us to selectively retrieve the text prompts most relevant to the bi-temporal hyperspectral images. The similarity computation is formulated as

$$\mathbf{S}^t_{dot} = \gamma\left(\mathbf{X}^t_{cls} \cdot \tilde{\mathbf{X}}_{text}\right) \tag{2}$$

where γ is a scaling factor used to scale the similarity.

We then sort $\mathbf{S}^{T1}_{dot}$ and $\mathbf{S}^{T2}_{dot}$ in descending order, respectively, and extract the top k dot-product scores along with corresponding text embeddings $\mathbf{X}^{T1}_{text}, \mathbf{X}^{T2}_{text} \in \mathbb{R}^{k\times d}$. These selected k dot-product scores and k text embeddings are subsequently used for feature fusion.

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After calculating the scaled dot-product similarity, we obtain the truncated bi-temporal text embeddings $\mathbf{X}^t_{\text{text}}$. To further evaluate the similarity between text embeddings and high-dimensional visual embeddings, we compute the Kullback–Leibler (KL) divergence, formulated as

$$\begin{cases} \mathbf{P}^t_{\text{img}} = \text{LogSoftmax}\left(\mathbf{X}^t_{\text{cls}}\right), & \mathbf{P}^t_{\text{img}} \in \mathbb{R}^{1\times C} \\ \mathbf{P}^t_{\text{text}} = \text{Softmax}\left(\mathbf{X}^t_{\text{text}}\right), & \mathbf{P}^t_{\text{text}} \in \mathbb{R}^{k\times d} \\ \mathbf{S}^t_{\text{KL}} = \sum_{i=1}^{d} \mathbf{P}^t_{\text{img}}(i) \log \dfrac{\mathbf{P}^t_{\text{img}}(i)}{\mathbf{P}^t_{\text{text}}(i)}, & \mathbf{S}^t_{\text{KL}} \in \mathbb{R}^{k} \end{cases} \tag{3}$$

where $\mathbf{P}^t_{\text{img}}$ and $\mathbf{P}^t_{\text{text}}$ are the probability distributions from the visual and text embeddings via LogSoftmax and Softmax, respectively. As noted, $d = C$, ensuring that the KL divergence $\mathbf{S}^t_{\text{KL}}$ can be computed element-wise between the visual distribution and each of the top-k selected text embeddings.

To fully exploit the contextual information from both visual and textual modalities for effective cross-modal fusion, we employ a cross-attention mechanism following the above computations. This mechanism fuses the high-dimensional visual embeddings $\mathbf{X}^5_{\text{vis}}$ with the truncated bi-temporal text embeddings, capturing complementary information across modalities and enriching the contextual representations for final fusion. The cross-attention computation is formulated as

$$\mathbf{X}^t_{\text{cross}} = \text{Softmax}\left(\frac{\mathbf{X}^t_{\text{text}} \cdot \left(\mathbf{X}^5_{\text{vis}}\right)^t}{\sqrt{d}}\right)\left(\mathbf{X}^5_{\text{vis}}\right)^t \tag{4}$$

where $\mathbf{X}^t_{\text{cross}} \in \mathbb{R}^{k\times d}$. In practice, the cross-attention mechanism is integrated with patch embedding, feed-forward networks, and residual connections to construct a cross transformer block. The resulting $\mathbf{X}^t_{\text{cross}}$ is used as the updated text embedding to repeatedly perform cross-attention operations with $(\mathbf{X}^5_{\text{vis}})^t$. This process is iterated 3 times to produce the final fused representation $\mathbf{X}^t_{\text{cross}}$.

We then perform a weighted fusion of the three branches as in (2)–(4). A residual connection with the selected text embeddings $\mathbf{X}^t_{\text{text}}$ is added to obtain the cross-modal features, which integrate semantic information from both modalities

$$\begin{aligned} W &= \alpha\text{Softmax}\left(\mathbf{S}^t_{\text{dot}}\right) + (1-\alpha)\text{Softmax}\left(-\mathbf{S}^t_{\text{KL}}\right) \\ \mathbf{X}^t_{\text{vt}} &= (1+W)\mathbf{X}^t_{\text{text}} + \mathbf{X}^t_{\text{cross}} \end{aligned} \tag{5}$$

where α is the gating coefficient controlling the weighting between the two similarity metrics. To ensure consistency with the similarity measure, we negate $\mathbf{S}^t_{\text{KL}}$ before computing the weights, so that larger weights correspond to higher similarity.

Finally, we compute a score map by multiplying $\mathbf{X}^t_{\text{vt}}$ with $(\mathbf{X}^5_{\text{vis}})^t$, and concatenate it with the visual features to enhance representation

$$\left(\mathbf{X}^5_{\text{vis}}\right)^t = \text{Concat}\left(\left(\mathbf{X}^5_{\text{vis}}\right)^t, \mathbf{X}^t_{\text{vt}} \times \left(\mathbf{X}^5_{\text{vis}}\right)^t\right). \tag{6}$$

By utilizing the SGFF module described above, we filter land-cover categories and fuse the textual semantic features of high-confidence categories with the visual features. We further refine deep visual features using a cross-modal score map. Subsequently, we compute differences on multiscale bi-temporal visual features to determine whether the land cover has changed.

D. Change-Aware Feature Enhancement

After SGFF, we obtain the cross-modal features $\mathbf{X}^t_{\text{vt}}$, along with the high-level visual feature maps $(\mathbf{X}^5_{\text{vis}})^t$ concatenated with the corresponding cross-modal score maps. To learn reliable change patterns from limited training samples, we design a multiperspective CAFE module. Under few-shot scenarios, change patterns are often incomplete and unstable, and a single difference measure is insufficient to capture the diverse spatial, semantic, and spectral characteristics of hyperspectral changes, making the learned representations less robust to distribution shift. To address this issue, CAFE adopts a multibranch design to capture complementary change features from multiple perspectives, and further uses the cross-modal features $\mathbf{X}^t_{\text{vt}}$ to adaptively reweight the enhanced representations, thereby emphasizing informative changes and improving robustness and generalization when the test distribution deviates from the limited training data.

We begin by combining $(\mathbf{X}^5_{\text{vis}})^t$ with low-level visual features to form multiscale feature sets $\mathbf{X}_{\text{vis}} = \{(\mathbf{X}^i_{\text{vis}})^t \mid t \in \{\text{T1}, \text{T2}\}, i \in \{1,2,3,4,5\}\}$ for multibranch processing. The details of each branch are described as follows.

1) Spatial Branch: This branch captures low-level texture differences by computing absolute differences between bi-temporal features at each scale, followed by a nonlinear transformation

$$\Delta\mathbf{F}^i_{\text{sub}} = \varphi\left(\left|\left(\mathbf{X}^i_{\text{vis}}\right)^{\text{T1}} - \left(\mathbf{X}^i_{\text{vis}}\right)^{\text{T2}}\right|\right), \quad i \in \{1,2,3,4,5\} \tag{7}$$

where $\varphi(\cdot)$ is a learnable nonlinear mapping. With the absolute difference, more similar feature pairs $(\mathbf{X}^i_{\text{vis}})^{\text{T1}}$ and $(\mathbf{X}^i_{\text{vis}})^{\text{T2}}$ yield smaller values of $\Delta\mathbf{F}^i_{\text{sub}}$.

2) Frequency Branch: To extract spectral difference features, we propose a Gaussian-based frequency subtraction module. The bi-temporal features $(\mathbf{X}^i_{\text{vis}})^{\text{T1}}, (\mathbf{X}^i_{\text{vis}})^{\text{T2}}$ are transformed into the frequency domain via 2-D discrete Fourier transform (DFT)

$$\left(\mathbf{X}^i_{\text{freq}}\right)^{\text{T1}} = \mathcal{F}\left(\left(\mathbf{X}^i_{\text{vis}}\right)^{\text{T1}}\right), \quad \left(\mathbf{X}^i_{\text{freq}}\right)^{\text{T2}} = \mathcal{F}\left(\left(\mathbf{X}^i_{\text{vis}}\right)^{\text{T2}}\right) \tag{8}$$

where $\mathcal{F}(\cdot)$ denotes the 2-D DFT applied to the spatial dimensions. A Gaussian-shaped low-pass filter $\mathcal{G} \in \mathbb{R}^{h\times w}$ is constructed as

$$\mathcal{G}(u,v) = \exp\left(-\frac{(u-h/2)^2 + (v-w/2)^2}{2\left(r \cdot \min\left(h/2, w/2\right)\right)^2}\right) \tag{9}$$

where r is a hyperparameter that controls the cutoff frequency of the filter. We set $r = 0.2$ in our experiments. The frequency maps are decomposed into low- and high-frequency components via element-wise multiplication

$$\begin{aligned} \left(\mathbf{X}^i_{\text{low}}\right)^t &= \left(\mathbf{X}^i_{\text{freq}}\right)^t \odot \mathcal{G} \\ \left(\mathbf{X}^i_{\text{high}}\right)^t &= \left(\mathbf{X}^i_{\text{freq}}\right)^t \odot (1-\mathcal{G}) \end{aligned}, \quad i \in \{1,2,3,4,5\} \tag{10}$$

where $\odot$ represents element-wise multiplication. The frequency-domain difference is then computed separately for the low- and high-frequency parts

$$\begin{aligned} \Delta\mathbf{F}^i_{\text{low}} &= \left|\left(\mathbf{X}^i_{\text{low}}\right)^{\text{T1}} - \left(\mathbf{X}^i_{\text{low}}\right)^{\text{T2}}\right| \\ \Delta\mathbf{F}^i_{\text{high}} &= \left|\left(\mathbf{X}^i_{\text{high}}\right)^{\text{T1}} - \left(\mathbf{X}^i_{\text{high}}\right)^{\text{T2}}\right|. \end{aligned} \tag{11}$$

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A channel-wise gating parameter $g \in [0,1]^C$ is introduced to adaptively fuse the two components

$$\Delta\mathbf{F}'^{i}_{\text{freq}} = g \cdot \Delta\mathbf{F}^{i}_{\text{low}} + (1-g) \cdot \Delta\mathbf{F}^{i}_{\text{high}}, \quad g = \sigma(\mathbf{W}) \tag{12}$$

where $\mathbf{W} \in \mathbb{R}^C$ is a learnable parameter and $\sigma(\cdot)$ is the sigmoid function. Finally, the fused frequency difference $\Delta\mathbf{F}'^{i}_{\text{freq}}$ is mapped back to the spatial domain via inverse DFT

$$\Delta\mathbf{F}^{i}_{\text{freq}} = \mathcal{F}^{-1}\left(\Delta\mathbf{F}'^{i}_{\text{freq}}\right) \tag{13}$$

where $\Delta\mathbf{F}^{i}_{\text{freq}}$ is the final result in the frequency branch. The frequency difference features corresponding to the multiscale features can be obtained.

3) Contextual Branch: This branch employs an FPN to extract difference features from multiscale representations, while simultaneously supplementing rich visual context information to support the other branches and enhance the accuracy of the subsequent classification head. Specifically, we first concatenate the bi-temporal features $\mathbf{X}_{\text{vis}} = \{(\mathbf{X}^{i}_{\text{vis}})^{t} \mid i \in \{1,2,3,4,5\}\}$ in a pairwise manner to form $\{(\mathbf{X}^{i}_{\text{vis}})^{\text{cat}}\}$. The FPN then processes these concatenated features to capture hierarchical differences, as defined below:

$$\Delta\mathbf{F}^{i}_{\text{fpn}} = \text{Conv}_{3\times3}\left(\text{Conv}_{1\times1}\left(\left(\mathbf{X}^{i}_{\text{vis}}\right)^{\text{cat}}\right) + \text{Upsample}\left(\text{Conv}_{1\times1}\left(\left(\mathbf{X}^{i+1}_{\text{vis}}\right)^{\text{cat}}\right)\right)\right). \tag{14}$$

4) Semantic Branch: This branch targets semantic-level changes. It computes the cosine similarity between bi-temporal features $\mathbf{X}_{\text{vis}} = \{(\mathbf{X}^{i}_{\text{vis}})^{t}\}$. Similar to the spatial branch, the cosine similarity is transformed via a sigmoid function, such that larger semantic discrepancies result in higher values of $\Delta\mathbf{F}_{\cos}$. For each bi-temporal feature pair $(\mathbf{X}^{i}_{\text{vis}})^{\text{T1}}$ and $(\mathbf{X}^{i}_{\text{vis}})^{\text{T2}}$, the transformation is defined as

$$\Delta\mathbf{F}^{i}_{\cos} = \sigma\left(1 - \frac{\left(\mathbf{X}^{i}_{\text{vis}}\right)^{\text{T1}} \cdot \left(\mathbf{X}^{i}_{\text{vis}}\right)^{\text{T2}}}{\left\|\left(\mathbf{X}^{i}_{\text{vis}}\right)^{\text{T1}}\right\|_2, \left\|\left(\mathbf{X}^{i}_{\text{vis}}\right)^{\text{T2}}\right\|_2}\right). \tag{15}$$

Given that cosine similarity predominantly captures high-level semantic differences and is less sensitive to low-level variations, we enhance $\Delta\mathbf{F}^{i}_{\cos}$ using the FPN-derived outputs $\Delta\mathbf{F}^{i}_{\text{fpn}}$

$$\Delta\mathbf{F}^{i}_{\cos} = \Delta\mathbf{F}^{i}_{\cos} \odot \Delta\mathbf{F}^{i}_{\text{fpn}}, \quad i \in \{1,2,3,4,5\}. \tag{16}$$

5) Multimodal-Guided Feature Attention: To summarize, the model extracts bi-temporal differences from four complementary perspectives: $\Delta\mathbf{F}^{i}_{\text{sub}}, \Delta\mathbf{F}^{i}_{\text{freq}}, \Delta\mathbf{F}^{i}_{\text{fpn}}, \Delta\mathbf{F}^{i}_{\cos}, i \in \{1,2,3,4,5\}$. Since change patterns vary across different regions, we perform a weighted fusion of these four perspectives. To this end, we introduce a text-guided channel attention mechanism that adaptively emphasizes informative features. Leveraging the cross-modal visual–text features $\mathbf{X}^{t}_{\text{vt}}$, we first concatenate the four types of difference features into $\Delta\mathbf{F}^{i}_{\text{all}} \in \mathbb{R}^{4C\times H\times W}$. Meanwhile, the bi-temporal $\mathbf{X}^{t}_{\text{vt}}$ are linearly projected into $4C$-dimensional space to obtain $\mathbf{X}'^{t}_{\text{vt}} \in \mathbb{R}^{k\times 4C}$. Cross-attention is then applied to produce a guided feature map $\Delta\mathbf{F}^{i}_{\text{vt}}$ that incorporates semantic priors

$$\Delta\mathbf{F}^{i}_{\text{vt}} = \text{Softmax}\left(\frac{\Delta\mathbf{F}^{i}_{\text{all}} \cdot \mathbf{X}'^{t}_{\text{vt}}}{\sqrt{4C}}\right) \cdot \mathbf{X}'^{t}_{\text{vt}}, \quad i \in \{1,2,3,4,5\}. \tag{17}$$

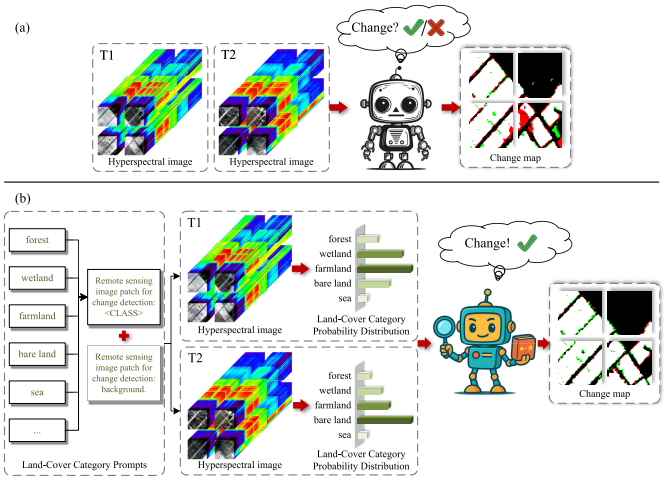


Fig. 1. Inference pathway of the proposed PromptHSI-CD is shown. By introducing predefined land-cover category prompts, the inference pathway is transformed from the traditional pixel-to-binary change map paradigm into a pixel-to-land-cover category probability distribution-to-binary change map paradigm. In the change map, black denotes unchanged regions, white denotes areas of change, green denotes FPs, and red denotes FNs. (a) Traditional HSI-CD. (b) PromptHSI-CD.

Following the SE block design in [47], the guided features $\Delta\mathbf{F}_{\text{vt}}$ undergo global pooling, followed by dimensionality reduction and expansion via a two-layer fully connected module with a reduction ratio of 16. The final sigmoid activation produces channel-wise weights to recalibrate $\Delta\mathbf{F}^{i}_{\text{all}}$

$$\Delta\mathbf{F}^{i}_{\text{change}} = \Delta\mathbf{F}^{i}_{\text{all}} + \Delta\mathbf{F}^{i}_{\text{all}} \odot \sigma\left(W_1\left(\eta\left(W_2 \cdot \Delta\mathbf{F}_{\text{vt}}\right)\right)\right) \tag{18}$$

where $\eta(\cdot)$ denotes the ReLU activation for introducing nonlinearity. The residual formulation stabilizes training and prevents over-amplification of activations.

E. Differential Feature Decoder

After the above steps, we obtain multiscale difference features $\Delta\mathbf{F}^{i}_{\text{change}}, i \in \{1,2,3,4,5\}$, which are semantically enhanced and sensitive to changes. Given that HSI-CD is performed at the patch level with focus on the central pixel, complex upsampling or decoding is unnecessary. A lightweight classification head is sufficient.

As the backbone is based on a Vision Transformer (ViT), all features share the same spatial resolution and channel dimension, allowing direct concatenation into $\Delta\mathbf{F} \in \mathbb{R}^{(5\times4C)\times H\times W}$. The classification head consists of convolutional layers with ReLU activations, whose depth is adaptively configured according to the feature map size to produce the final logit $Y \in \mathbb{R}^{2\times1\times1}$. A Softmax layer then outputs the binary classification result.

In summary, PromptHSI-CD introduces predefined land-cover semantics through multimodal feature extraction and establishes a semantically guided inference pathway from pixels to the final change map. By incorporating transferable category-level semantic information, it reduces reliance on low-level spectral change patterns that are difficult to learn robustly from limited labeled samples. The SGFF module builds a stable semantic intermediate representation, thereby mitigating overfitting under few-shot settings. The CAFE

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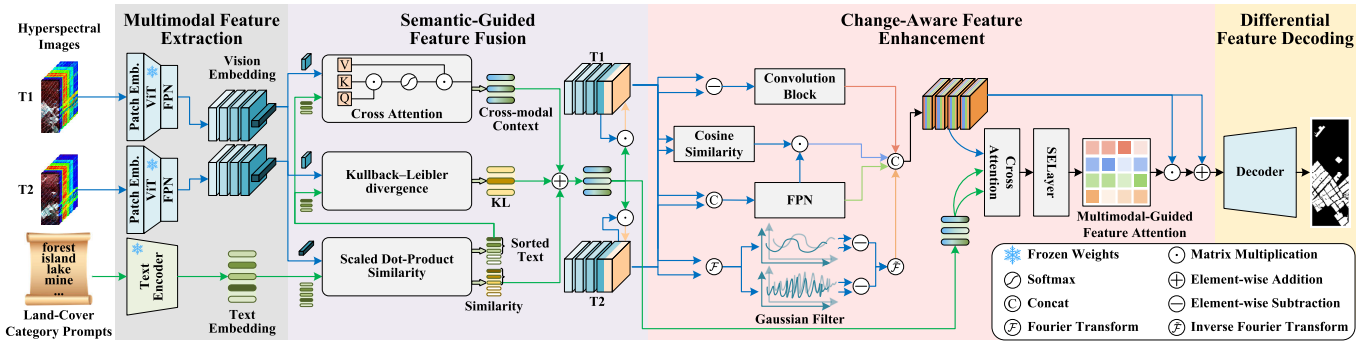


Fig. 2. Overall architecture of PromptHSI-CD, including a multimodal feature extraction backbone, an SGFF module, a CAFE module, and a differential feature decoder.

TABLE II
SUMMARY OF THE FIVE HSI-CD DATASETS USED IN THE EXPERIMENTS

Datasets	Sensor	Location	Time (T1 / T2)	Image Size (H / W / C)	Resolution (Spat. / Spec.)	Main Change Category
Chinese Farmland	Hyperion	Jiangsu, China	2006 / 2007	420 / 140 / 154	30 m / 10 nm	Farmland
American Farmland	Hyperion	Oregon, USA	2004 / 2007	307 / 241 / 159	30 m / 10 nm	Irrigation
River	Hyperion	Jiangsu, China	2013-05 / 2013-12	463 / 241 / 198	30 m / 10 nm	Suspended substances
Santa Barbara	AVIRIS	California, USA	2013 / 2014	984 / 740 / 224	15.2 m / 10 nm	Cropland
Bay Area	AVIRIS	California, USA	2013 / 2015	600 / 500 / 224	16.9 m / 10 nm	Cropland

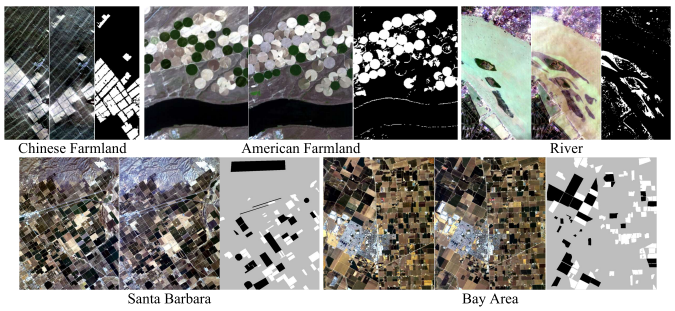


Fig. 3. False-color display of bi-temporal images and ground truth of five datasets. In the ground truth, black indicates unchanged regions, white denotes areas of change, and gray denotes uncertain regions.

module further enhances difference features through multiperspective change modeling and multimodal-guided reweighting, enabling more robust extraction of complementary change features. Finally, the enhanced difference features are decoded to obtain the binary change map. As a result, change decisions depend not only on visual differences but also on whether the land-cover category has changed, improving robustness and generalization in few-shot scenarios.

IV. EXPERIMENTS

A. Datasets

To comprehensively evaluate the effectiveness of the proposed PromptHSI-CD, we conduct experiments on five widely used HSI-CD datasets: Chinese Farmland [48], American Farmland [48], River [17], Santa Barbara [49], and Bay Area [49]. Fig. 3 presents false-color bi-temporal image pairs together with the corresponding ground-truth (GT) maps for all benchmark datasets. Table II summarizes detailed information on each dataset.

B. Implementation Details

PromptHSI-CD initializes both the text and visual encoders with the pretrained CLIP ViT-B/16 weights. We use focal loss [50] with $\alpha = 0.25$ and $\gamma = 2$ and optimize the network with Adam starting from a learning rate of 1×10^{-4}, decayed by a cosine annealing schedule. We train for 100 epochs with a batch size of 64. Because the datasets differ in land-cover complexity and in the number of changed land-cover categories, the hyperparameter k, which controls how many land-cover prompts are retained, is tuned per dataset: 10 for American Farmland and Santa Barbara, 14 for Bay Area, and 15 for Chinese Farmland and River. A detailed analysis of the impact of different values of k on model performance is provided in Section IV-G2. All experiments are implemented in PyTorch and are run on NVIDIA L40S GPUs.

Following previous studies [36], [51], [52], [53], our preprocessing pipeline includes four steps: normalization, training/testing sample selection, padding, and patch extraction. Unlike most existing methods, we do not perform dimensionality reduction on the bi-temporal hyperspectral images to preserve the structural properties of the spectral information. As a result, the input channel number of the patch embedding layer in the visual encoder is dataset-dependent. Although this introduces the original band number as a hyperparameter, it eliminates the need to manually choose a reduced dimension and therefore requires no additional tuning.

Specifically, we first apply min–max normalization to $[0, 1]$ along the channel dimension for the bi-temporal hyperspectral images to reduce inter-dataset statistical differences. We then adopt a stratified sampling strategy to maintain a consistent class distribution between the training and testing sets. Specifically, 1% of the changed pixels and 1% of the unchanged pixels are randomly selected as training samples, and all the

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TABLE III
STATISTICS OF THE 1% SAMPLE SPLIT USED IN THE COMPARISON AND ABLATION EXPERIMENTS

Datasets	Train Set			Test Set			Uncertain Pixels	Total
	Changed Pixels	Unchanged Pixels	Total	Changed Pixels	Unchanged Pixels	Total		
Chinese Farmland	183	404	587	18200	40013	58213	0	58800
American Farmland	166	573	739	16510	56738	73248	0	73987
River	96	1018	1114	9602	100867	110469	0	111583
Santa Barbara	521	804	1325	51613	79614	131227	595608	728160
Bay Area	392	342	734	38878	33869	72747	226519	300000

remaining labeled pixels are used for testing. Uncertain pixels in the Santa Barbara and Bay Area datasets are unlabeled and thus excluded from sampling. Detailed sampling results are reported in Table III. Next, mirror padding with a width of half the patch size is used to enable edge pixels to serve as patch centers. Finally, patches are cropped around sampled pixels. To account for different spatial resolutions, the patch size is set to 5×5 for the coarse-resolution datasets (Chinese Farmland, American Farmland, and River) and 15×15 for the fine-resolution datasets (Santa Barbara and Bay Area).

C. Comparison Methods and Evaluation Metrics

To comprehensively evaluate the proposed PromptHSI-CD, we compare it with SOTA methods in the HSI-CD domain, including traditional unsupervised methods CVA [12] and PCAKM [13], as well as deep-learning-based approaches such as CLIP [29], ML-EDAN [54], GlobalMind [55], DIEFEN [53], MSCSCNet [56], STMNet [37], DADSSFF [36], BTCDNet [51], GPSST [57], and CGMNet [58]. For all comparison methods, we adopt the optimal hyperparameter settings and preprocessing procedures reported in their original articles, except that the training/testing sample split is unified across methods for a fair comparison. The results of STMNet are taken directly from its publication, whereas the results of all other methods are reproduced by us based on their original articles and official code. Since CVA and PCAKM are unsupervised methods, they do not require training samples. MSCSCNet, as a semi-supervised method, is trained with 1% labeled samples and 19% unlabeled samples. All other deep learning methods use the same training and testing partitions shown in Table III.

To evaluate the performance of PromptHSI-CD and ensure consistency with existing work [52], [53], [59], [60], we use common change detection metrics: overall accuracy (OA), Kappa coefficient, precision, recall, and $F1$-score. HSI-CD is formulated as a binary classification task that determines whether each pixel has changed or not. Therefore, its performance can be evaluated using the confusion matrix, which consists of true positives (TPs), true negatives (TNs), false positives (FPs), and false negatives (FNs). Based on these quantities, the following metrics are defined:

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{19}$$

$$\mathrm{Precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{20}$$

TABLE IV
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE CHINESE FARMLAND DATASET

Method	OA	Kappa	F1	Precision	Recall
CVA [12]	0.8794	0.7106	0.7957	0.8459	0.7511
PCAKM [13]	0.8695	0.6878	0.7803	0.8238	0.7411
CLIP [29]	0.9464	0.8755	0.9145	0.9119	0.9171
ML-EDAN [54]	0.9485	0.8800	0.9175	0.9182	0.9169
GlobalMind [55]	0.9493	0.8824	0.9194	0.9150	0.9237
DIEFEN [53]	0.9442	0.8714	0.9122	0.8978	0.9271
MSCSCNet [56]	0.9534	0.8900	0.9235	**0.9492**	0.8991
DADSSFF [36]	0.9322	0.8387	0.8870	0.9255	0.8516
BTCDNet [51]	0.9546	0.8943	0.9273	0.9296	0.9249
GPSST [57]	0.9558	0.8965	0.9286	0.9377	0.9197
CGMNet [58]	0.9435	0.8673	0.9081	0.9237	0.8930
PromptHSI-CD (ours)	**0.9614**	**0.9099**	**0.9380**	0.9420	**0.9340**

$$\mathrm{Recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{21}$$

$$\mathrm{F1} = \frac{2 \cdot \mathrm{Precision} \cdot \mathrm{Recall}}{\mathrm{Precision}+\mathrm{Recall}} \tag{22}$$

$$\mathrm{Kappa} = \frac{P_o - P_e}{1 - P_e} \tag{23}$$

$$P_o = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN}} \tag{24}$$

$$P_e = \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})+(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2}. \tag{25}$$

Due to the inherent class imbalance in HSI-CD, where unchanged pixels usually dominate the image, OA alone may not fully reflect the model performance. Therefore, we emphasize comprehensive metrics such as the $F1$-score and the Kappa coefficient, which provide a more reliable evaluation under class-imbalanced conditions.

D. Comparison With SOTA Methods

1) Chinese Farmland Dataset: Table IV reports the quantitative results on the Chinese Farmland dataset. Compared with GPSST, which is pretrained on hyperspectral data, PromptHSI-CD achieves better performance on all metrics. This indicates that, even without hyperspectral pretraining, PromptHSI-CD can still achieve strong performance with the guidance of land-cover category prompts. Fig. 4 presents qualitative visualizations, where white, black, green, and red denote TPs, TNs, FPs, and FNs, respectively. Traditional threshold- or clustering-based methods, such as CVA and PCAKM, rely

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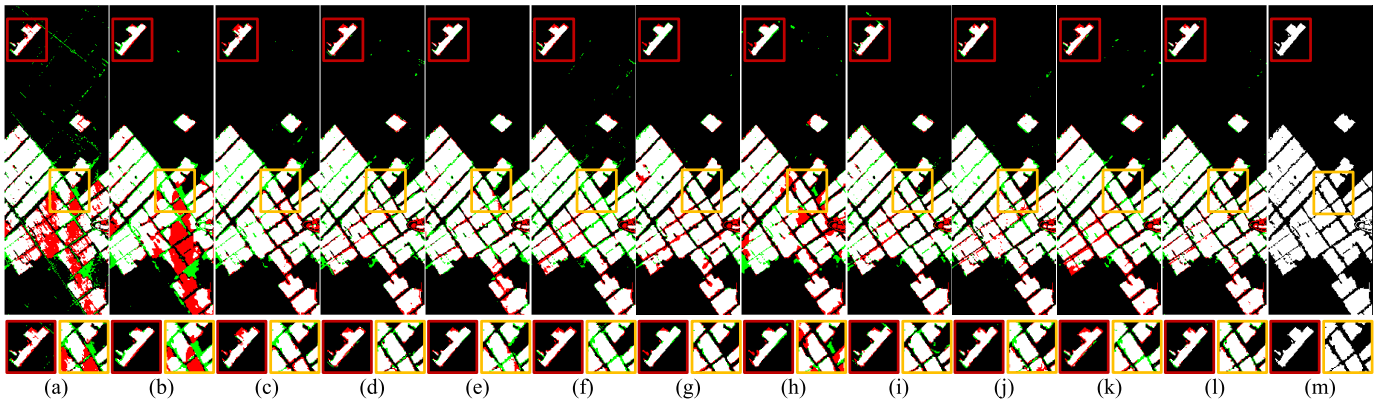


Fig. 4. Visualization of prediction results from different methods on the Chinese Farmland dataset: (a) CVA [12], (b) PCAKM [13], (c) CLIP [29], (d) ML-EDAN [54], (e) GlobalMind [55], (f) DIEFEN [53], (g) MSCSCNet [56], (h) DADSSFF [36], (i) BTCDNet [51], (j) GPSST [57], (k) CGMNet [58], (l) PromptHSI-CD (ours), and (m) ground truth. Black, white, green, and red denote unchanged regions, changed regions, FPs, and FNs, respectively.

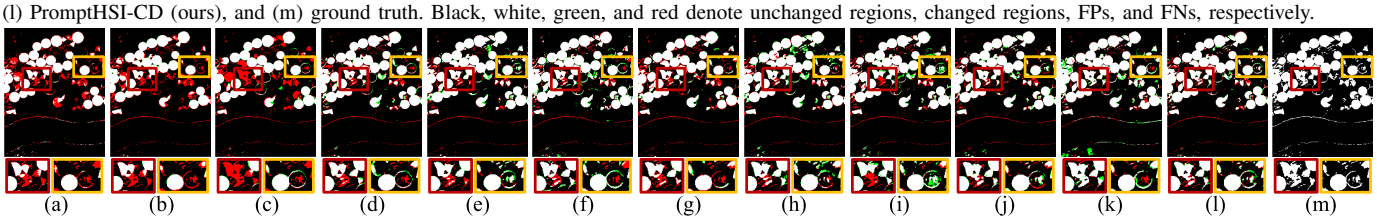


Fig. 5. Visualization of prediction results from different methods on the American farmland dataset: (a) CVA [12], (b) PCAKM [13], (c) CLIP [29], (d) ML-EDAN [54], (e) GlobalMind [55], (f) DIEFEN [53], (g) MSCSCNet [56], (h) DADSSFF [36], (i) BTCDNet [51], (j) GPSST [57], (k) CGMNet [58], (l) PromptHSI-CD (ours), and (m) ground truth. Black, white, green, and red denote unchanged regions, changed regions, FPs, and FNs, respectively.

TABLE V
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE AMERICAN FARMLAND DATASET

Method	OA	Kappa	F1	Precision	Recall
CVA [12]	0.9290	0.7714	0.8135	**0.9963**	0.6873
PCAKM [13]	0.9307	0.7797	0.8213	0.9813	0.7062
CLIP [29]	0.9171	0.7344	0.7837	0.9517	0.6661
ML-EDAN [54]	0.9516	0.8552	0.8858	0.9464	0.8325
GlobalMind [55]	0.9577	0.8755	0.9024	0.9396	0.8681
DIEFEN [53]	0.9568	0.8737	0.9013	0.9290	0.8753
MSCSCNet [56]	0.9464	0.8367	0.8701	0.9596	0.7958
STMNet [37]	0.9514	0.8571	0.8881	0.8564	**0.9223**
DADSSFF [36]	0.9586	0.8796	0.9062	0.9259	0.8873
BTCDNet [51]	0.9533	0.8622	0.8920	0.9310	0.8561
GPSST [57]	0.9487	0.8479	0.8805	0.9266	0.8386
CGMNet [58]	0.9581	0.8800	0.9070	0.9083	0.9057
PromptHSI-CD (ours)	**0.9620**	**0.8879**	**0.9121**	0.9517	0.8757

solely on local spectral change values and are easily affected by natural environmental factors such as illumination, resulting in a large number of FP and FN errors. Among the deep learning methods, ML-EDAN, GlobalMind, DIEFEN, GPSST, and CGMNet tend to misclassify unchanged roads as changed, especially in the yellow-boxed region where farmland and roads are interlaced, whereas MSCSCNet, DADSSFF, and BTCDNet often fail to preserve farmland boundaries and produce varying degrees of FN errors. In contrast, PromptHSI-CD accurately delineates the irregular farmland in the upper-left region and clearly distinguishes changed farmland from unchanged roads. This superiority can be attributed to its use of land-cover category semantics to determine whether category transitions occur over time.

2) American Farmland Dataset: Table V summarizes the quantitative results on the American Farmland dataset. Compared with the SOTA fully supervised method CGMNet, PromptHSI-CD achieves superior performance on all metrics except recall. This demonstrates that the proposed design can effectively cope with complex change patterns and deliver strong detection performance. Fig. 5 provides qualitative comparisons. In this scene, the main changed regions are relatively evident, but there are also small changed regions, as shown in the yellow-boxed area. Among the comparison methods, only GlobalMind, DADSSFF, BTCDNet, and GPSST can detect a limited number of changed pixels in the center of this ring-shaped region, but they still produce many FP and FN errors in the surrounding ring-shaped area. In contrast, PromptHSI-CD more accurately identifies the changes in this small region and preserves a more complete delineation of the surrounding ring-shaped structure. In the left enlarged view, PromptHSI-CD also shows fewer misses in the crescent-shaped change area, yielding a more continuous change mask. These gains are attributed to a multiperspective change modeling strategy, which adaptively emphasizes the most informative perspective and thus enables the model to handle diverse and complex change patterns, rather than being limited to the dominant crop-type shifts or moisture variations in this dataset.

3) River Dataset: Table VI presents the quantitative results on the River dataset. It is evident that PromptHSI-CD achieves a more effective balance between precision and recall than the other deep learning methods. This is mainly because the River dataset suffers from severe class imbalance. As shown in Table III, the training set contains only 96 changed samples, making it difficult to effectively learn the diverse change patterns in the data. In contrast, PromptHSI-CD

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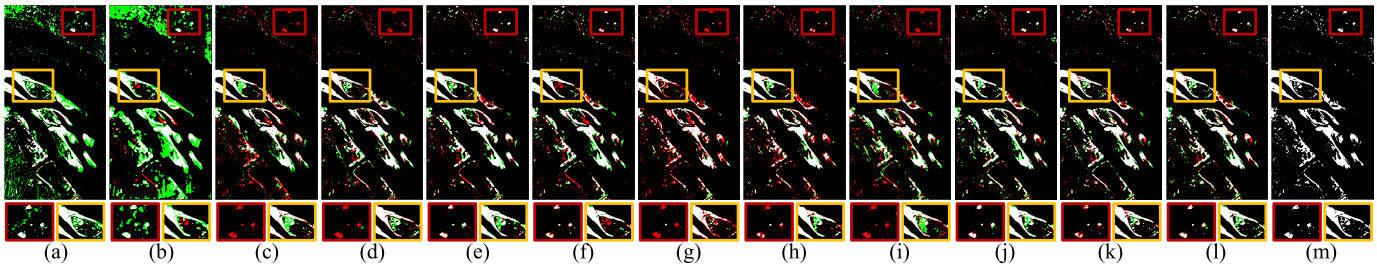


Fig. 6. Visualization of prediction results from different methods on the River dataset: (a) CVA [12], (b) PCAKM [13], (c) CLIP [29], (d) ML-EDAN [54], (e) GlobalMind [55], (f) DIEFEN [53], (g) MSCSCNet [56], (h) DADSSFF [36], (i) BTCDNet [51], (j) GPSST [57], (k) CGMNet [58], (l) PromptHSI-CD (ours), and (m) ground truth. Black, white, green, and red denote unchanged regions, changed regions, FPs, and FNs, respectively.

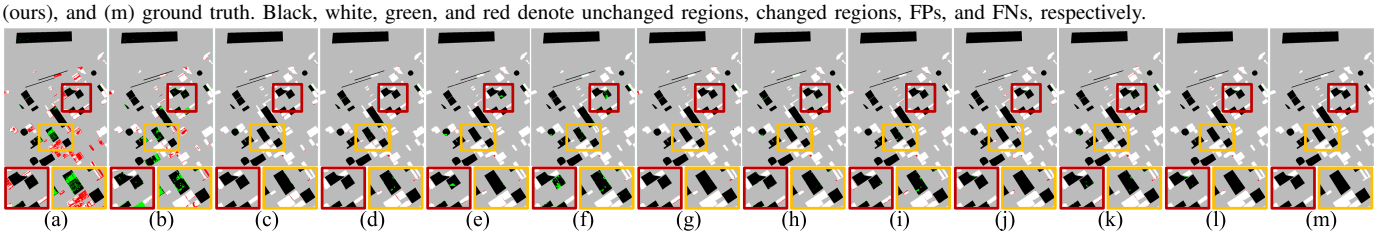


Fig. 7. Visualization of prediction results from different methods on the Santa Barbara dataset: (a) CVA [12], (b) PCAKM [13], (c) CLIP [29], (d) ML-EDAN [54], (e) GlobalMind [55], (f) DIEFEN [53], (g) MSCSCNet [56], (h) DADSSFF [36], (i) BTCDNet [51], (j) GPSST [57], (k) CGMNet [58], (l) PromptHSI-CD (ours), and (m) ground truth. Black, white, gray, green, and red denote unchanged regions, changed regions, uncertain regions, FPs, and FNs, respectively.

TABLE VI
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE RIVER DATASET

Method	OA	Kappa	F1	Precision	Recall
CVA [12]	0.9275	0.6593	0.6970	0.5475	**0.9588**
PCAKM [13]	0.8661	0.4697	0.5340	0.3827	0.8829
CLIP [29]	0.9507	0.6698	0.6965	0.7495	0.6505
ML-EDAN [54]	0.9637	0.7645	0.7843	0.8100	0.7602
GlobalMind [55]	0.9645	0.7713	0.7907	0.8103	0.7719
DIEFEN [53]	0.9635	0.7666	0.7866	0.8009	0.7728
MSCSCNet [56]	0.9570	0.6729	0.6946	0.9089	0.5621
DADSSFF [36]	0.9621	0.7354	0.7557	**0.8588**	0.6747
BTCDNet [51]	0.9507	0.6805	0.7074	0.7306	0.6856
GPSST [57]	0.9645	0.7735	0.7930	0.8032	0.7830
CGMNet [58]	0.9671	0.7866	0.8045	0.8307	0.7799
PromptHSI-CD (ours)	**0.9674**	**0.7925**	**0.8103**	0.8195	0.8014

TABLE VII
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE SANTA BARBARA DATASET

Method	OA	Kappa	F1	Precision	Recall
CVA [12]	0.8231	0.6073	0.7325	0.9038	0.6158
PCAKM [13]	0.9037	0.7989	0.8787	0.8708	0.8867
CLIP [29]	0.9901	0.9791	0.9873	**0.9938**	0.9808
ML-EDAN [54]	0.9861	0.9708	0.9822	0.9891	0.9754
GlobalMind [55]	0.9866	0.9720	0.9830	0.9846	0.9814
DIEFEN [53]	0.9788	0.9560	0.9737	0.9531	**0.9952**
MSCSCNet [56]	0.9910	0.9812	0.9886	0.9863	0.9908
DADSSFF [36]	0.9865	0.9718	0.9829	0.9807	0.9852
BTCDNet [51]	0.9835	0.9653	0.9789	0.9851	0.9728
GPSST [57]	0.9896	0.9783	0.9869	0.9810	0.9929
CGMNet [58]	0.9826	0.9635	0.9778	0.9810	0.9746
PromptHSI-CD (ours)	**0.9926**	**0.9845**	**0.9906**	0.9915	0.9897

maintains balanced precision and recall under such extreme data scarcity, achieves the best OA, Kappa, and F1, and attains the highest recall among all deep learning methods. As shown in Fig. 6, traditional methods perform poorly because large illumination differences between the bi-temporal images alter the spectral responses in regions where the land cover actually remains unchanged, leading to many FP errors. Deep learning methods also show limited ability to detect fine-grained changes. Among the comparison methods, only GlobalMind, DIEFEN, and GPSST can completely detect the three changed areas within the red-boxed region. Overall, ML-EDAN, GlobalMind, DIEFEN, GPSST, and CGMNet achieve performance close to that of our method; however, both quantitative and qualitative analyses show that PromptHSI-CD still maintains a leading performance. PromptHSI-CD not only captures the fine-grained change patches, but also more accurately detects the large-scale river-bank changes in the yellow-boxed region and the lower-middle part of the image. This advantage is mainly attributed to the predefined land-cover category prompts, which provide semantic guidance for change detection and reduce the reliance on large-scale labeled data. In addition, the pixel-level guidance enhances land-cover discrimination and improves sensitivity to small-object changes.

4) Santa Barbara Dataset: Table VII presents the quantitative results on the Santa Barbara dataset. Compared with the semi-supervised method MSCSCNet, PromptHSI-CD achieves higher OA, Kappa, F1, and precision, while showing only a marginal difference in recall. Considering that MSCSCNet additionally uses unlabeled samples during training, these results further verify the effectiveness of the proposed semantic-guided multimodal method. Fig. 7 provides qualitative comparisons. Although the visual differences among the deep learning methods are relatively small, PromptHSI-CD shows clearer boundary delineation and fewer FP or FN errors in the boxed regions. Moreover, outside these regions, PromptHSI-CD does not exhibit concentrated error areas, such as the left-edge

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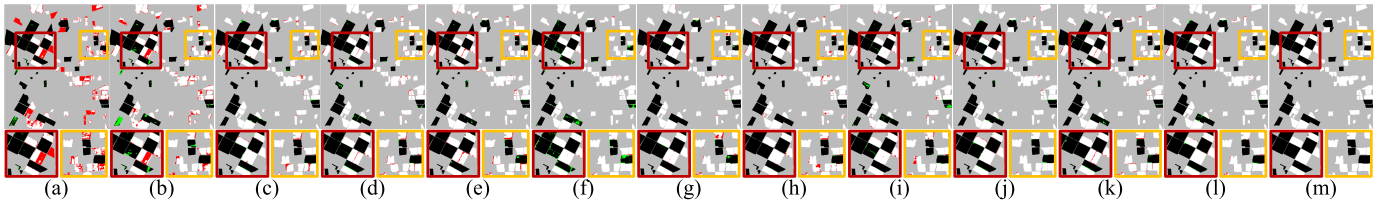


Fig. 8. Visualization of prediction results from different methods on the Bay Area dataset: (a) CVA [12], (b) PCAKM [13], (c) CLIP [29], (d) ML-EDAN [54], (e) GlobalMind [55], (f) DIEFEN [53], (g) MSCSCNet [56], (h) DADSSFF [36], (i) BTCDNet [51], (j) GPSST [57], (k) CGMNet [58], (l) PromptHSI-CD (ours), and (m) ground truth. Black, white, gray, green, and red denote unchanged regions, changed regions, uncertain regions, FPs, and FNs, respectively.

TABLE VIII
QUANTITATIVE EVALUATION RESULTS OF DIFFERENT METHODS ON THE BAY AREA DATASET

Method	OA	Kappa	F1	Precision	Recall
CVA [12]	0.8003	0.6082	0.7762	0.9677	0.6479
PCAKM [13]	0.8856	0.7719	0.8865	0.9429	0.8365
CLIP [29]	0.9703	0.9404	0.9720	0.9807	0.9634
ML-EDAN [54]	0.9708	0.9415	0.9725	0.9805	0.9646
GlobalMind [55]	0.9722	0.9442	0.9737	0.9862	0.9614
DIEFEN [53]	0.9672	0.9338	0.9699	0.9495	**0.9913**
MSCSCNet [56]	0.9836	0.9672	0.9846	0.9893	0.9800
STMNet [37]	0.9138	0.8256	0.9225	0.8940	0.9528
DADSSFF [36]	0.9778	0.9555	0.9789	**0.9942**	0.9640
BTCDNet [51]	0.9635	0.9267	0.9657	0.9693	0.9622
GPSST [57]	0.9831	0.9661	0.9843	0.9813	0.9873
CGMNet [58]	0.9771	0.9540	0.9784	0.9857	0.9713
PromptHSI-CD (ours)	**0.9838**	**0.9675**	**0.9849**	0.9857	0.9841

errors in MSCSCNet and the errors within the lower-left circular regions in GPSST and CGMNet. Since the main changes in this dataset are crop variations with distinct spectral differences, most HSI-CD methods achieve relatively strong performance. Nevertheless, methods incorporating textual information, such as CLIP and PromptHSI-CD, still outperform the other deep learning baselines, demonstrating the benefit of introducing a text branch. Compared with CLIP, PromptHSI-CD further improves intraclass consistency and suppresses FP through the SGFF and CAFE modules, while its multiperspective modeling of spatial, frequency, and semantic differences enables more accurate delineation of change boundaries, leading to SOTA performance.

5) Bay Area Dataset: Table VIII reports the quantitative results on the Bay Area dataset. Compared with MSCSCNet trained with additional unlabeled samples, PromptHSI-CD still achieves the highest OA, Kappa, and F1 on the Bay Area dataset. This further demonstrates the effectiveness of the proposed semantic-guided multimodal modeling strategy. Fig. 8 shows qualitative comparisons. In the two boxed views, compared with the semi-supervised method MSCSCNet, the pretraining-based method GPSST, and the recent fully supervised method CGMNet, PromptHSI-CD produces fewer errors along field boundaries and achieves better intrafield consistency, without concentrated FP or FN errors. This gain stems in part from the CAFE module, which captures change features in the frequency domain and strengthens fine-grained edge delineation compared with models operating solely in the spatial domain. In addition, illumination and shadow differences between the two acquisitions often lead to the same object exhibiting different spectra, which makes purely spectral methods prone to FP. By introducing fine-grained textual prompts, our method associates pixel spectra with predefined land-cover categories, mitigates error propagation due to spectral inconsistency of unchanged surfaces, and focuses the decision on whether the land-cover category has truly changed.

6) Joint Analysis of the Five Datasets: Across all five datasets, PromptHSI-CD consistently achieves leading OA, Kappa, and $F1$-scores, while maintaining balanced precision and recall even under extreme data scarcity and severe class imbalance. From the perspective of spatial resolution, the datasets are divided into two groups: the Chinese Farmland, American Farmland, and River datasets acquired by the Hyperion sensor with a spatial resolution of 30 m, and the Santa Barbara and Bay Area datasets acquired by the AVIRIS sensor with an approximate spatial resolution of 16 m. For each compared deep learning method, we computed the average performance on the 30- and 16-m groups, as well as the performance gap between the two resolutions. The results are reported in Table IX. As shown, all methods achieve better performance on the fine-resolution (16 m) datasets than on the coarse-resolution (30 m) datasets. This is mainly because land-cover mixing is more severe at coarse resolution, which weakens the spectral contrast between changed and unchanged areas, whereas fine-resolution pixels usually correspond to more homogeneous land-cover types and thus provide clearer change cues. The Δ columns report the average performance gaps across resolutions; smaller values indicate lower sensitivity to spatial resolution variations. Among the compared methods, DIEFEN shows relatively low sensitivity to spatial resolution, while PromptHSI-CD achieves the smallest gap in F1 and the second smallest gap in Kappa. This can be attributed to the introduced land-cover category prompts, which provide stable semantic guidance, and the proposed CAFE module, which enhances change representation in a multiscale and multiperspective manner. Considering both performance and sensitivity, PromptHSI-CD achieves favorable detection results while exhibiting relatively low sensitivity to spatial resolution variations, which indicates its potential robustness and generalizability.

E. Few-Shot Experimental Analysis

In this section, we evaluate the performance of PromptHSI-CD under different levels of training data availability. Following the sampling protocol described in Section IV-B, we adopt a stratified sampling strategy for

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TABLE IX
AVERAGE PERFORMANCE OF DEEP LEARNING METHODS AT DIFFERENT SPATIAL RESOLUTIONS AND THE AVERAGE PERFORMANCE GAPS ACROSS RESOLUTIONS

Method	OA			Kappa			F1		
	30m	16m	Δ	30m	16m	Δ	30m	16m	Δ
CLIP [29]	0.9381	0.9802	0.0421	0.7599	0.9598	0.1999	0.7982	0.9796	0.1814
ML-EDAN [54]	0.9546	0.9785	0.0239	0.8332	0.9561	0.1229	0.8625	0.9774	0.1148
GlobalMind [55]	0.9572	0.9794	0.0223	0.8431	0.9581	0.1151	0.8708	0.9783	0.1075
DIEFEN [53]	0.9549	0.9730	**0.0181**	0.8372	0.9449	**0.1077**	0.8667	0.9718	0.1051
MSCSCNet [56]	0.9523	0.9873	0.0350	0.7999	0.9742	0.1743	0.8294	0.9866	0.1572
DADSSFF [36]	0.9510	0.9822	0.0312	0.8179	0.9636	0.1457	0.8496	0.9809	0.1313
BTCDNet [51]	0.9529	0.9735	0.0206	0.8123	0.9460	0.1337	0.8422	0.9723	0.1301
GPSST [57]	0.9563	0.9864	0.0301	0.8393	0.9722	0.1329	0.8673	0.9856	0.1183
CGMNet [58]	0.9562	0.9798	0.0236	0.8446	0.9588	0.1141	0.8732	0.9781	0.1049
PromptHSI-CD (ours)	**0.9636**	**0.9882**	0.0246	**0.8634**	**0.9760**	0.1126	**0.8868**	**0.9877**	**0.1009**

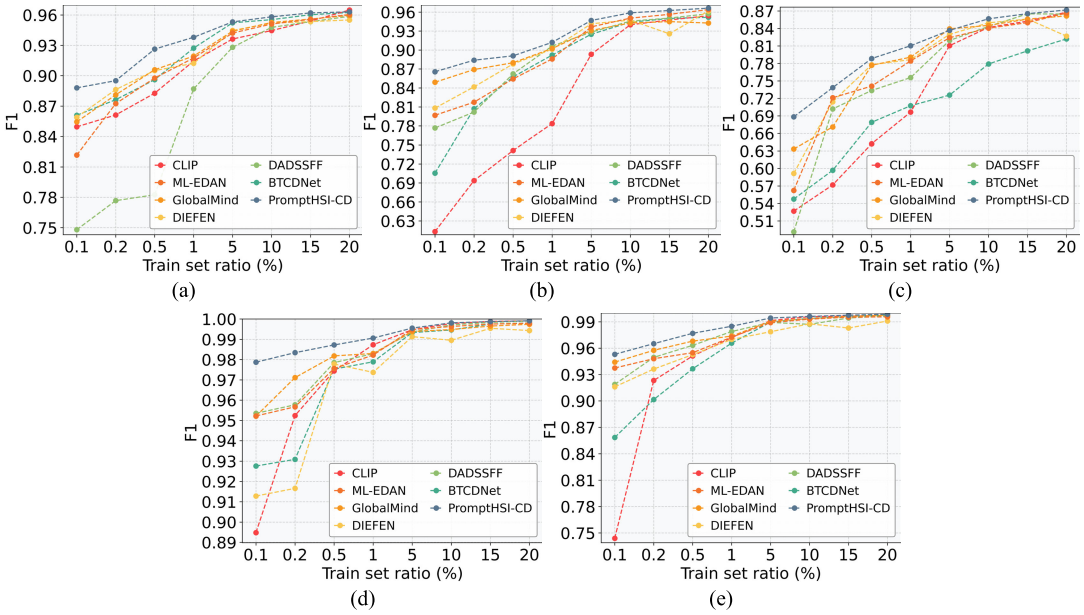


Fig. 9. Performance comparison of different methods and PromptHSI-CD under varying training sample ratios across the five datasets. (a) Chinese Farmland, (b) American Farmland, (c) River, (d) Santa Barbara, and (e) Bay Area.

each dataset by separately sampling changed and unchanged pixels. Specifically, we consider eight sampling ratios: 0.1%, 0.2%, 0.5%, 1%, 5%, 10%, 15%, and 20%. For example, for the Chinese Farmland dataset at a sampling ratio of 0.1%, 0.1% of the changed pixels and 0.1% of the unchanged pixels are selected as training samples. Referring to the 1% sampling statistics in Table III, this corresponds to 18 changed samples and 40 unchanged samples, yielding a total of 58 training samples, while the remaining 58 742 samples are used for testing. The test results of all models under the eight sampling ratios are presented in Fig. 9.

With only 0.1% of the training data, PromptHSI-CD outperforms all competing methods on every dataset. A clear trend is observed as the training ratio decreases: PromptHSI-CD exhibits a slower degradation in performance, which is especially evident on Chinese Farmland and Santa Barbara. This advantage arises because, under limited supervision, conventional deep learning models that map pixels directly to a binary change map struggle to learn generalizable spectral representations. In contrast, PromptHSI-CD leverages predefined textual prompts to supply additional semantics and transforms the inference pathway from pixel-to-binary into pixel-to-land-cover category probability distribution-to-binary. This allows the model to infer land-cover categories first and then decide whether change has occurred; the pixel-to-land-cover category probability mapping is semantically richer and less sensitive to illumination and other disturbance factors. In addition, the SGFF module aligns and fuses visual features with textual prompts at a fine granularity using attention and KL divergence, and the CAFE module captures change features from multiple perspectives to improve adaptation to diverse change patterns. These designs enhance robustness and generalization in low-data regimes, enabling reliable change detection when training samples are scarce.

F. Ablation Study

To verify the effectiveness of the proposed SGFF and CAFE modules, we construct several representative model variants by decoupling individual components and conduct corresponding

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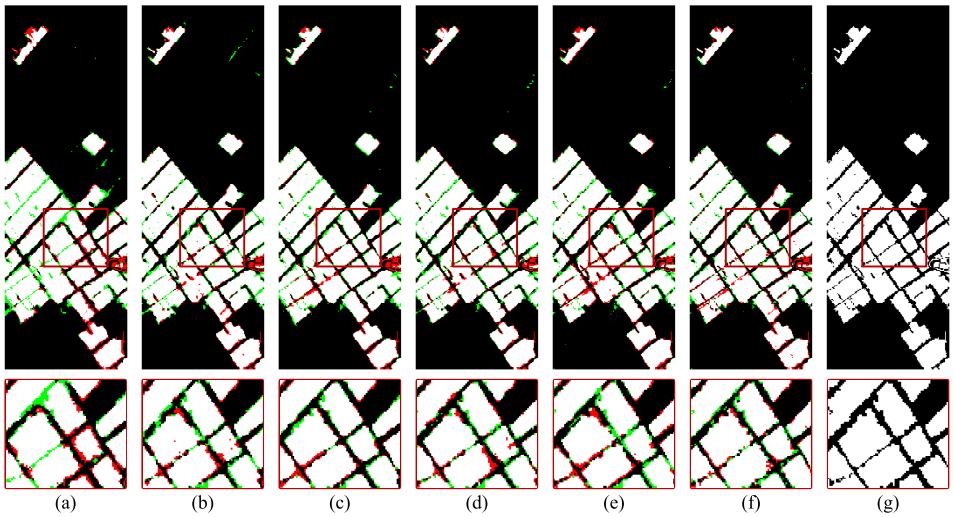


Fig. 10. Visualization of prediction results from different ablation settings on the Chinese Farmland dataset: (a) baseline, (b) baseline + Sub, (c) baseline + Cat, (d) baseline + CAFE, (e) baseline + SGFF + CAFE, (f) PromptHSI-CD, and (g) ground truth. Black, white, green, and red denote unchanged regions, changed regions, FPs, and FNs, respectively.

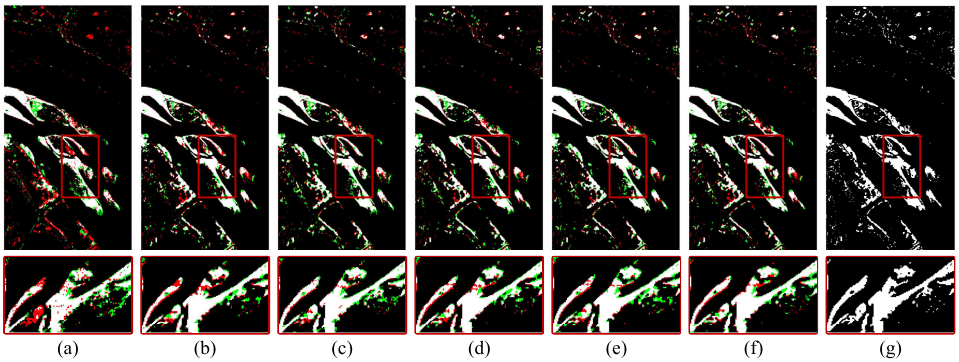


Fig. 11. Visualization of prediction results from different ablation settings on the River dataset: (a) baseline, (b) baseline + Sub, (c) baseline + Cat, (d) baseline + CAFE, (e) baseline + SGFF + CAFE, (f) PromptHSI-CD, and (g) ground truth. Black, white, green, and red denote unchanged regions, changed regions, FPs, and FNs, respectively.

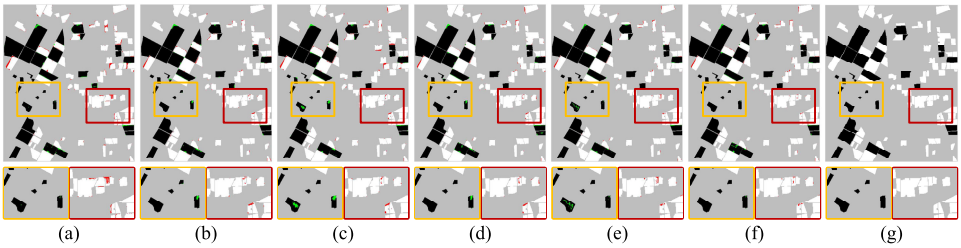


Fig. 12. Visualization of prediction results from different ablation settings on the Bay Area dataset: (a) baseline, (b) baseline + Sub, (c) baseline + Cat, (d) baseline + CAFE, (e) baseline + SGFF + CAFE, (f) PromptHSI-CD, and (g) ground truth. Black, white, gray, green, and red denote unchanged regions, changed regions, uncertain regions, FPs, and FNs, respectively.

ablation experiments. The adopted baseline is a CLIP model adapted for HSI-CD, where the original multimodal encoders are used to extract features, and the [CLS] tokens from both modalities are fed into a linear decoder to generate the change map. To examine the contribution of the CAFE module, we replace it with subtraction (baseline + Sub) and concatenation (baseline + Cat) operations for difference feature extraction. In particular, to exclude the influence of insufficient fusion between visual and textual modalities, we use only the visual branch in these variants. Thus, in baseline + CAFE, the CAFE module excludes the cross-modal attention component and focuses solely on whether multiperspective change-feature enhancement can effectively improve model performance. We then introduce the text branch and the SGFF module to achieve cross-modal feature fusion. In baseline + SGFF + CAFE, the CAFE module still omits the cross-modal attention mechanism, retaining only the SGFF-driven enhancement of deep features to verify the effectiveness of SGFF itself. Finally, PromptHSI-CD represents the complete proposed method, which incorporates cross-modal attention guided by textual semantics to highlight the contribution of multimodal features in enhancing change representations.

Table X reports the quantitative results of the above variants. As shown in the table, the multiperspective strategy

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TABLE X

BASELINE: CLIP ENCODERS WITHOUT FEATURE FUSION OR ENHANCEMENT; SUB: SUBTRACTION-BASED CHANGE-FEATURE ENHANCEMENT; CAT: CONCATENATION-BASED CHANGE-FEATURE ENHANCEMENT; CAFE (W/O ATT): CAFE MODULE WITHOUT MULTIMODAL-GUIDED FEATURE ATTENTION; PROMPTHSI-CD: FULL FRAMEWORK WITH SGFF, CAFE, AND ATTENTION MECHANISM

Dataset	Model	baseline	baseline + Sub	baseline + Cat	baseline + CAFE	baseline + SGFF + CAFE	PromptHSI-CD
	Encoder	Image / Text	Image	Image	Image	Image / Text	Image / Text
	Fusion	-	-	-	-	SGFF	SGFF
	Enhancement	-	Subtraction	Concatenate	CAFE(w/o att)	CAFE(w/o att)	CAFE
	Attention	-	-	-	-	-	✓
	Decoder	linear	Conv + linear	Conv + linear	Conv + linear	Conv + linear	Conv + linear
Chinese Farmland	OA	0.9464	0.9589	0.9585	0.9601	0.9558	**0.9614**
	Kappa	0.8755	0.9046	0.9038	0.9076	0.8969	**0.9099**
	F1	0.9145	0.9346	0.9340	0.9367	0.9290	**0.9380**
	Precision	0.9119	0.9296	0.9283	0.9290	0.9336	**0.9420**
	Recall	0.9171	0.9395	0.9399	**0.9445**	0.9244	0.9340
River	OA	0.9507	0.9651	0.9626	0.9646	0.9651	**0.9674**
	Kappa	0.6698	0.7740	0.7672	0.7777	0.7821	**0.7925**
	F1	0.6965	0.7930	0.7877	0.7971	0.8012	**0.8103**
	Precision	0.7495	0.8184	0.7766	0.7941	0.7939	**0.8195**
	Recall	0.6505	0.7692	0.7992	0.8001	**0.8086**	0.8014
Bay Area	OA	0.9703	0.9815	0.9735	0.9807	0.9836	**0.9838**
	Kappa	0.9404	0.9628	0.9467	0.9612	0.9671	**0.9675**
	F1	0.9720	0.9826	0.9751	0.9819	0.9846	**0.9849**
	Precision	0.9807	0.9838	0.9791	0.9812	**0.9857**	**0.9857**
	Recall	0.9634	0.9815	0.9711	0.9827	0.9835	**0.9841**

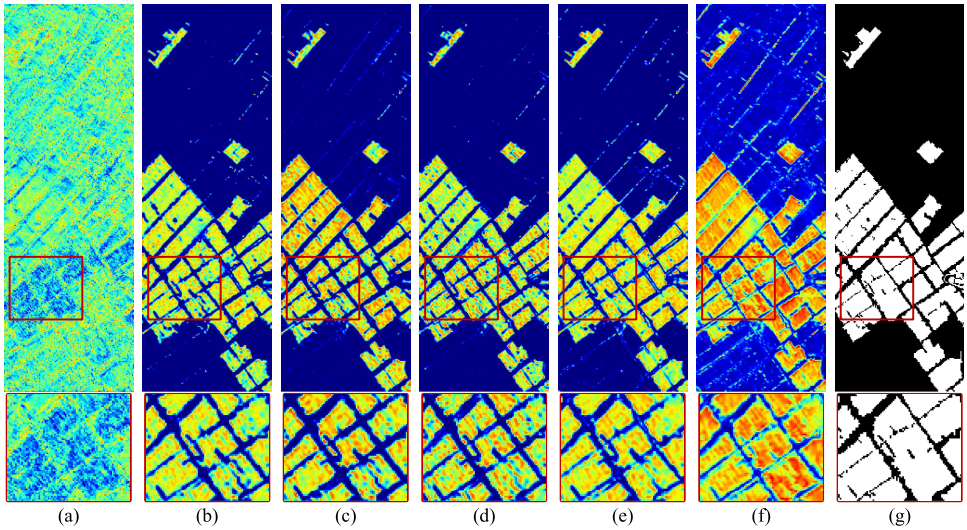


Fig. 13. Visualization of the feature heatmaps fed into the decoder under different ablation settings on the Chinese Farmland dataset: (a) baseline, (b) baseline + Sub, (c) baseline + Cat, (d) baseline + CAFE, (e) baseline + SGFF + CAFE, (f) PromptHSI-CD, and (g) ground truth.

introduced by baseline + CAFE outperforms the single subtraction or concatenation operations on the Chinese Farmland and River datasets with more complex change patterns. The visual comparisons in Figs. 10–12 further support this observation: in Fig. (d) (baseline + CAFE), green FPs are markedly reduced relative to Fig. (b) and (c). This improvement mainly stems from the change-feature enhancement across spatial, frequency, and semantic views, which equips the model to handle diverse change patterns. The frequency-domain branch also helps the model capture clearer land-cover transitions, yielding sharper change boundaries and fewer red FN along edges. On the simpler Bay Area dataset, baseline + CAFE remains competitive. After introducing the text modality to refine deep features, baseline + SGFF + CAFE achieves a notable performance gain, demonstrating the feasibility of using predefined land-cover textual prompts to guide the model. However, because baseline + SGFF + CAFE only refines deep features with text and does not apply attention weighting to the multiperspective change features, it surpasses baseline + CAFE only on River and Bay Area. By contrast, the full PromptHSI-CD, which incorporates cross-modal attention to weight change features under multimodal feature guidance, can adaptively adjust the contributions of different perspectives according to various change patterns, exhibiting better generalization and thus achieving the highest performance across all datasets.

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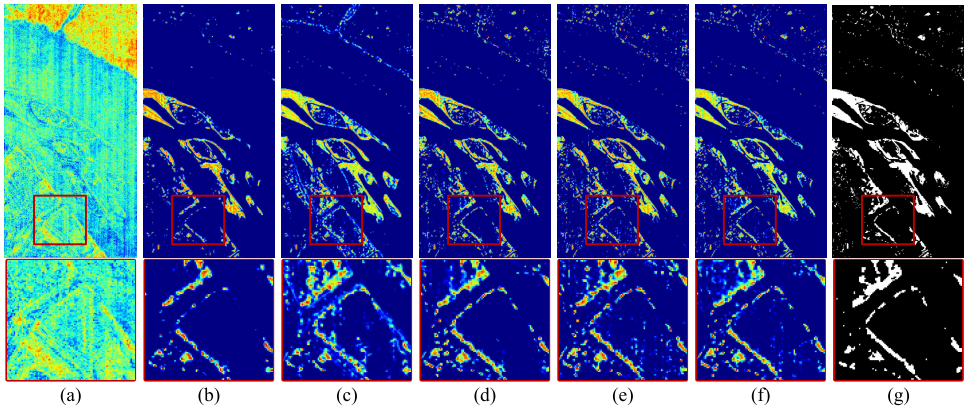


Fig. 14. Visualization of the feature heatmaps fed into the decoder under different ablation settings on the River dataset: (a) baseline, (b) baseline + Sub, (c) baseline + Cat, (d) baseline + CAFE, (e) baseline + SGFF + CAFE, (f) PromptHSI-CD, and (g) ground truth.

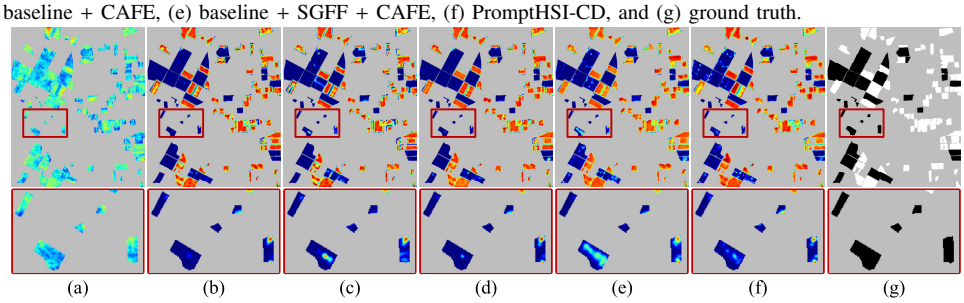


Fig. 15. Visualization of the feature heatmaps fed into the decoder under different ablation settings on the Bay Area dataset: (a) baseline, (b) baseline + Sub, (c) baseline + Cat, (d) baseline + CAFE, (e) baseline + SGFF + CAFE, (f) PromptHSI-CD, and (g) ground truth.

Figs. 13–15 visualize the Grad-CAM heatmaps of the features fed into the decoder for different model variants. For the baseline in Fig. 13(a), where no change-feature enhancement is applied, the responses over changed and unchanged regions are not clearly distinguishable, indicating that the model attends almost uniformly to different land-cover categories. In Fig. 13, the complete model in Fig. 13(f) exhibits much stronger responses over the true change regions: large areas of changed farmland are highlighted in red and orange. The zoomed-in region at the bottom further shows that, compared with other variants, the full model better handles the intricate patterns where farmland and narrow roads are intertwined. This improvement mainly benefits from the fine-grained textual prompts and the cross-modal attention mechanism, which together enable adaptive selection of the most informative enhancement branches and highlight more complete and detailed change patterns. In Fig. 14, the zoomed-in region at the bottom shows that Fig. 14(b), which relies solely on subtraction, captures only coarse spatial structures and misses many subtle sediment changes along the river. Variant Fig. 14(c) with concatenation focuses on redundant features and fails to emphasize the main change areas. In contrast, variant Fig. 14(d) with CAFE delineates the change boundaries more clearly and preserves small change regions. Building on this, variants Fig. 14(e) and (f) further incorporate the text modality to refine deep features and introduce attention guidance, leading to more focused responses over the change regions. In Fig. 15, the zoomed-in region illustrates the ability of our method to suppress regions prone to FP. Although the subtraction-based variant Fig. 15(b) shows limited FP in the zoomed-in region, this is mainly due to its weak capability to capture changes, which is also evident in the red zoomed region in Fig. 12. Compared with Fig. 12(c), which uses concatenation, Fig. 12(d) with CAFE more reliably identifies the area as unchanged by aggregating information from multiple perspectives. After introducing textual information to refine deep features, variant Fig. 12(e) becomes more sensitive to land-cover changes, which may introduce local FP. However, with the subsequent cross-modal attention in variant Fig. 12(f), the multiperspective features are reweighted, effectively suppressing these local FP and demonstrating that the full model generalizes well across different change patterns.

G. Additional Analysis

In this section, we analyze the encoder freezing strategy and the choice of the hyperparameter k, and discuss the parameter scale and inference time of the compared deep learning models and PromptHSI-CD.

1) Encoder Freezing Strategy: Since the original CLIP model is pretrained on RGB images, we freeze the encoders when transferring it to hyperspectral images in order to preserve the strong textual semantic extraction capability of the CLIP text encoder while better isolating the contribution of the proposed SGFF and CAFE modules. To examine this design choice, we conduct an ablation study under four settings: only the text encoder is frozen, only the vision encoder is frozen, both encoders are unfrozen, and both encoders are frozen. As reported in Table XI, freezing both encoders yields the best OA, Kappa, and $F1$-scores on the River and Santa Barbara datasets, ranks second on the Chinese Farmland and Bay Area datasets, and shows relatively weaker performance only on the American Farmland dataset. In terms of average performance over the five datasets, freezing both encoders gives

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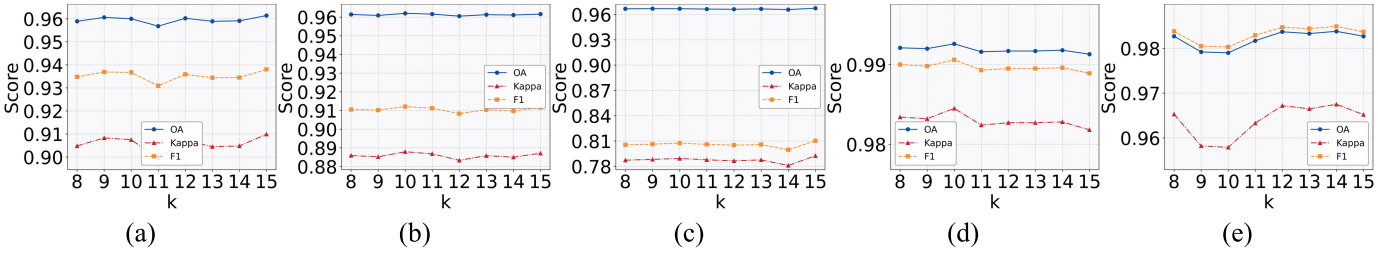


Fig. 16. Performance analysis of PromptHSI-CD with different k values on the five datasets. (a) Chinese Farmland, (b) American Farmland, (c) River, (d) Santa Barbara, and (e) Bay Area.

TABLE XI

QUANTITATIVE COMPARISON OF DIFFERENT VARIANTS WITH FROZEN OR UNFROZEN ENCODERS. ✓ DENOTES FROZEN (NONTRAINABLE), AND ✗ DENOTES UNFROZEN (TRAINABLE)

Datasets	Text Encoder	Vision Encoder	OA	Kappa	F1	Precision	Recall
Chinese Farmland	✓	✗	0.9573	0.9005	0.9316	0.9327	0.9304
	✗	✓	0.9605	0.9080	0.9368	0.9361	0.9375
	✗	✗	**0.9620**	**0.9115**	**0.9392**	0.9384	**0.9401**
	✓	✓	0.9614	0.9099	0.9380	**0.9420**	0.9340
American Farmland	✓	✗	0.9619	0.8875	0.9118	0.9521	0.8747
	✗	✓	0.9628	0.8899	0.9136	**0.9577**	0.8733
	✗	✗	**0.9640**	**0.8939**	**0.9168**	0.9573	**0.8795**
	✓	✓	0.9620	0.8879	0.9121	0.9517	0.8757
River	✓	✗	0.9658	0.7822	0.8009	0.8103	0.7917
	✗	✓	0.9658	0.7819	0.8007	0.8108	0.7908
	✗	✗	0.9637	0.7730	0.7929	0.7860	0.7999
	✓	✓	**0.9674**	**0.7925**	**0.8103**	**0.8195**	**0.8014**
Santa Barbara	✓	✗	0.9902	0.9795	0.9876	0.9874	0.9877
	✗	✓	0.9922	0.9836	0.9901	0.9881	**0.9920**
	✗	✗	0.9893	0.9777	0.9865	0.9866	0.9864
	✓	✓	**0.9926**	**0.9845**	**0.9906**	**0.9915**	0.9897
Bay Area	✓	✗	0.9782	0.9561	0.9796	0.9789	0.9803
	✗	✓	**0.9840**	**0.9679**	**0.9851**	0.9849	**0.9852**
	✗	✗	0.9778	0.9555	0.9791	0.9851	0.9732
	✓	✓	0.9838	0.9675	0.9849	**0.9857**	0.9841
Average	✓	✗	0.9707	0.9012	0.9223	0.9323	0.9130
	✗	✓	0.9730	0.9063	0.9252	0.9355	0.9158
	✗	✗	0.9714	0.9023	0.9229	0.9307	0.9158
	✓	✓	**0.9734**	**0.9085**	**0.9272**	**0.9381**	**0.9169**

slightly more favorable results across all metrics. Overall, the differences between freezing and unfreezing the encoders are not large. Although unfreezing the encoders can lead to slightly better performance on some individual datasets, the fully frozen setting appears to provide somewhat more stable results under the 1% training-sample condition. A possible reason is that the limited labeled samples are insufficient to represent the full data distribution, so updating a larger number of encoder parameters may fit dataset-specific noise rather than more transferable change patterns.

Therefore, after jointly considering the limited supervision condition and the cross-dataset generalizability, we adopt the strategy of freezing the encoders and training only the subsequent modules. Under our experimental setting, whether the encoders are frozen or not does lead to substantial differences, suggesting that the RGB-to-HSI modality gap can be partially alleviated by the subsequent SGFF and CAFE modules.

TABLE XII

QUANTITATIVE COMPARISON BETWEEN THE OPTIMALLY FILTERED k TEXT PROMPTS AND THE FULL 56-CLASS TEXT PROMPTS $k = 56$

Dataset	k	OA	Kappa	F1	Precision	Recall
Chinese Farmland	15	**0.9614**	**0.9099**	**0.9380**	**0.9420**	**0.9340**
	56	0.9561	0.8980	0.9300	0.9262	**0.9340**
American Farmland	10	**0.9620**	**0.8879**	**0.9121**	0.9517	**0.8757**
	56	0.9618	0.8868	0.9111	**0.9583**	0.8683
River	15	**0.9674**	**0.7925**	**0.8103**	0.8195	**0.8014**
	56	0.9667	0.7859	0.8042	**0.8220**	0.7870
Santa Barbara	10	**0.9926**	**0.9845**	**0.9906**	0.9915	**0.9897**
	56	0.9921	0.9835	0.9900	**0.9917**	0.9882
Bay Area	14	**0.9838**	**0.9675**	**0.9849**	**0.9857**	**0.9841**
	56	0.9803	0.9604	0.9815	0.9846	0.9784

2) Hyperparameter k: We conduct parameter experiments on the hyperparameter k, which controls the number of selected text prompts. When k is too small, the model relies excessively on the encoder's modality alignment capability, which may lead to incorrect guidance for subsequent feature fusion. Conversely, an excessively large k introduces redundant textual prompts, reducing the discriminability of bi-temporal features and weakening the model's ability to capture true change regions. Therefore, we vary k within the range of 8–15 for evaluation. As shown in Fig. 16, the impact of k on performance is minor for the relatively simple Santa Barbara dataset, whereas for datasets with more diverse land-cover categories such as Chinese Farmland and American Farmland, the choice of k has a more noticeable effect. Based on the overall results, the hyperparameter k is tuned per dataset: 10 for American Farmland and Santa Barbara, 15 for Chinese Farmland and River, and 14 for Bay Area.

Furthermore, to examine whether the Top-k truncation strategy improves model performance, we compare the optimal k values for each dataset with the case of $k = 56$, where all land-cover categories are retained. The results, shown in Table XII, indicate that using Top-k truncation consistently yields better overall performance across all datasets, with only a slight decrease in precision in a few cases compared with $k = 56$. This is mainly because a large number of textual prompts blur the distinction between bi-temporal features, making the model more inclined to predict unchanged regions. As a result, FPs are reduced, while FNs increase, leading to higher precision but lower recall and lower F1.

3) Parameter Scale and Inference Time: Table XIII presents the number of trainable parameters and the inference time of

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TABLE XIII
COMPARISON OF PARAMETERS AND INFERENCE TIME ACROSS DIFFERENT METHODS

Method	Params (M)	Inference Time (s)
CLIP [29]	201.39	0.67
ML-EDAN [54]	112.28	**0.19**
GlobalMind [55]	2.73	-
DIEFEN [53]	8.07	0.54
STMNet [37]	44.90	-
DADSSFF [36]	5.13	0.22
BTCDNet [51]	**1.55**	0.38
PromptHSI-CD (ours)	110.24	2.44

different deep learning models on a simulated hyperspectral dataset with a spatial size of 175 × 175 and 154 spectral channels. Since GlobalMind [55] processes the entire image in a single forward pass rather than performing pixel-wise inference in batches, and STMNet [37] operates on single-temporal images only, their inference times are not directly comparable to the other methods. Compared with other deep learning methods, PromptHSI-CD contains a larger number of trainable parameters, mainly because the introduction of the textual branch increases both the parameter count and computational cost, resulting in a longer inference time. However, the additional parameters enable the model to better capture cross-modal correlations, thereby improving detection performance. Moreover, PromptHSI-CD is not a simple accumulation of parameters. Compared with larger models such as CLIP and ML-EDAN, it achieves superior detection accuracy while requiring far less labeled data, reaching SOTA performance with only 0.1% of the training samples.

V. DISCUSSION

Although PromptHSI-CD achieves strong performance across multiple hyperspectral datasets, several limitations remain. Its effectiveness depends on predefined land-cover prompts, which may be less suitable when change categories are ambiguous, incomplete, or weakly aligned with scene content. In addition, the model still requires dataset-dependent settings such as patch size, suggesting that its adaptability to variations in spatial resolution and scene complexity is not yet fully automatic. As a multimodal method, it also has a higher computational cost than simpler baselines, which may limit its efficiency in resource-constrained applications.

In future work, we will explore more suitable strategies for generating textual prompts tailored to hyperspectral images, enabling more targeted land-cover guidance. We will also investigate lightweight multimodal solutions for HSI-CD to further improve inference efficiency. In addition, extending the framework to semi-supervised or even unsupervised learning settings is a promising direction for further reducing the reliance on labeled data.

VI. CONCLUSION

We proposed PromptHSI-CD, a multimodal method for HSI-CD that leveraged predefined land-cover category prompts to guide HSI-CD. Through the SGFF module and CAFE module, the model utilized textual prompts to refine deep visual features, enhanced change representations from multiple perspectives, and employed cross-modal feature guidance to direct attention, enabling accurate detection with limited training samples. Experiments on five datasets showed that PromptHSI-CD outperformed other baselines across various metrics. Even with only 0.1% labeled data, the model maintained strong performance and generalization. Ablation studies confirmed the importance of predefined land-cover category prompts, the SGFF module, and the CAFE module. This work demonstrated the potential of language-vision fusion in remote sensing and offered new directions for multimodal HSI-CD.

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